

Mohism Lab

Dr Qingbiao Li
Mohism Lab

Assistant Prof at Dept of EME and Adjunct Prof at Dept of CIS at FST from University of Macau

Associate Researcher at Zhuhai UM Science & Technology Research Institute

Founding member of UM-PolyU Joint Laboratory of Robotics and Embodied Intelligence

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Homepage : <https://qingbiaoli.github.io>



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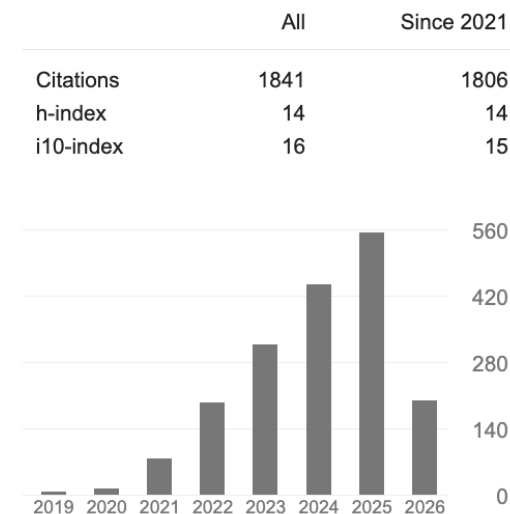
About me

Work Experience

08.2024 - Present	Assistant Professor at University of Macau
01.2023 – 04.2024	Postdoctoral Researcher for RAILS (Self-Driving Car)
	Line manager: Dr Lars Kunze
07.2021 – 11.2021	Research Intern at Microsoft Cambridge Research
	Line manager: Dr Ioan.stefanovici
09.2016 – 06.2017	Research Assistant in Bipedal Walking of Humanoid Robot
	Supervisors: Dr Zhibin Li & Prof Sethu Vijayakumar

Education Background

10.2018 – 12.2022	PhD in Computer Science (Multi-agent/robot motion planning)
	Supervisor: Prof Amanda Prorok at Prorok Lab, University of Cambridge
10.2017 – 10.2018	MRes Medical Robotics and Image Guided Intervention
	Supervisor: Prof Daniel Elson at Hamlyn Centre, Imperial College London
09.2013 - 06.2016	M. Eng (Hons) Mechanical Engineering
	Supervisor: Dr Filipe Teixeira-Dias at The University of Edinburgh
09.2011 - 07.2013	B. Eng Mechanical Engineering and Automation
	South China University of Technology



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My Research journey

Received the 2021 Publication of the Year in Computer Science and Technology from the University of Cambridge.

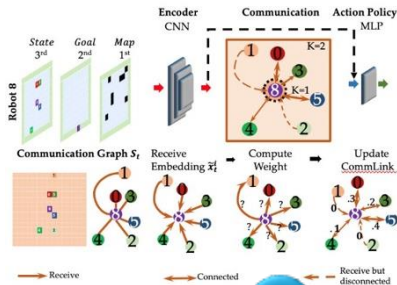


2021



A surgical scene recognition and surgical field enhancement technology based on generative adversarial networks (ICRA 2021)

Graph Neural Network for Multi-robot Path Planning (IROS 2020, RAL 2021, TSP2022, SSRR 2022)

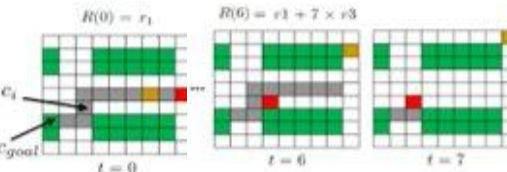


2022

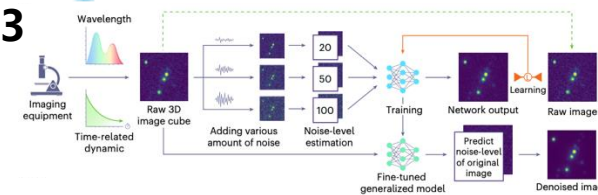


A Framework for Real-World Multi-Robot Systems Running Decentralized GNN-Based Policies (ICRA 2021, ICRA 2022)

Machine learning algorithm that introduces human prior knowledge (RAL 2020, ICLR 2022, ICRA 2023)

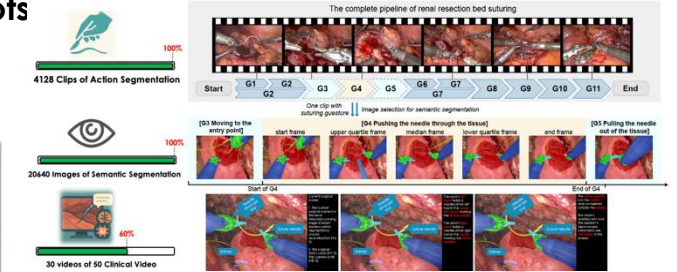


2023

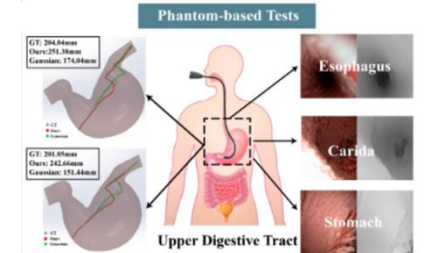


Self-supervised Deep Learning for Tracking Degradation of Perovskite Light-emitting Diodes with Multispectral Imaging (Nature Machine Intelligence 2023)sc

Automated Surgery (da Vinci Surgical Robot) – Real Clinical Dataset SRA-RPN40 (ICRA 2026 , and Ongoing)



2025



Depth Estimation and Localization of Monocular Camera Based on Graph Neural Network in Endoscopic Surgery (JBHI 2025) Automated surgery using Notes (IROS 2025)



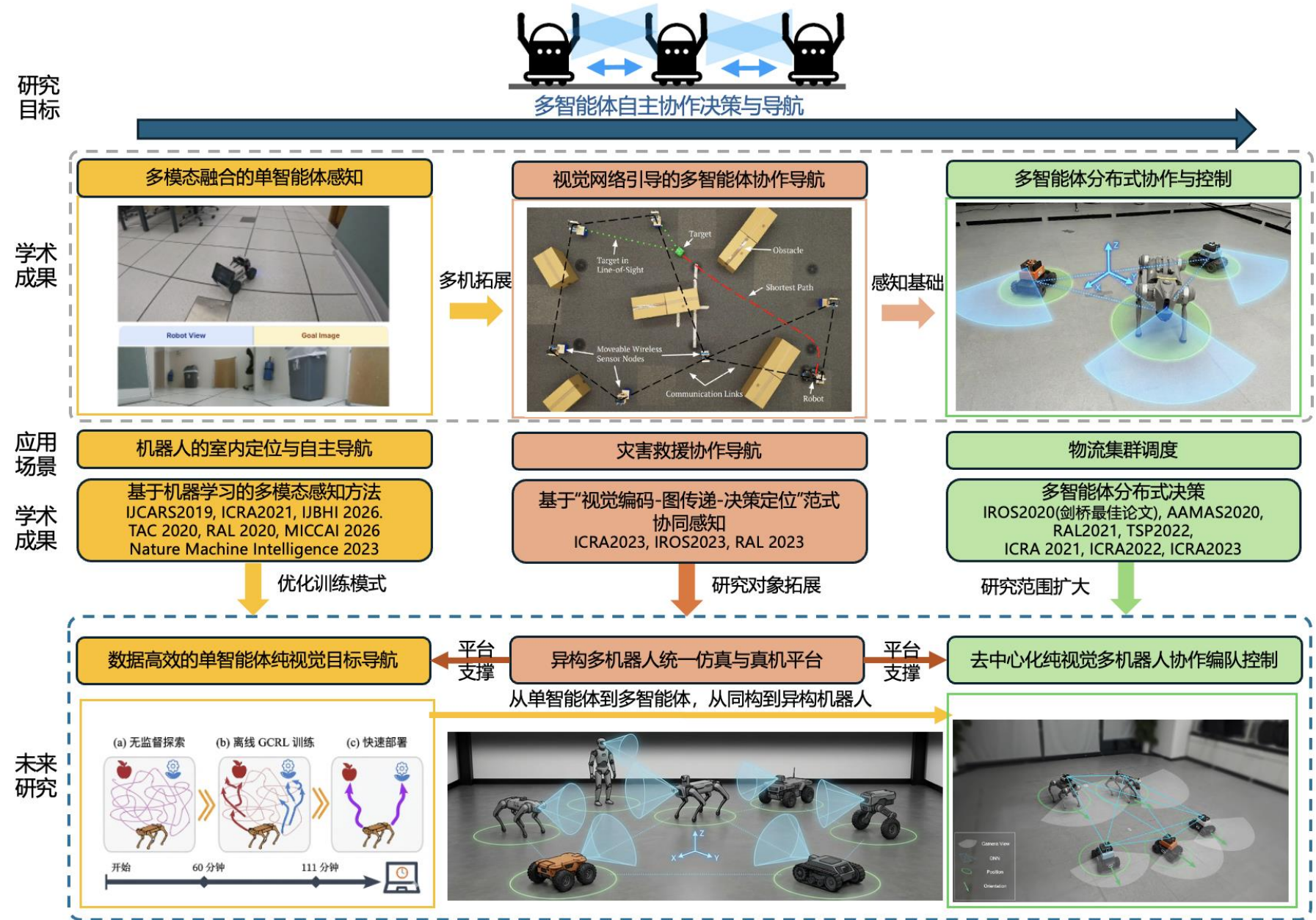
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Research Focus: Machine Learning for Embodied Artificial Intelligence

➤ **Research Direction:** Robot Learning for multi-robot and multi agent collaboration, LLMs-based autonomous agent



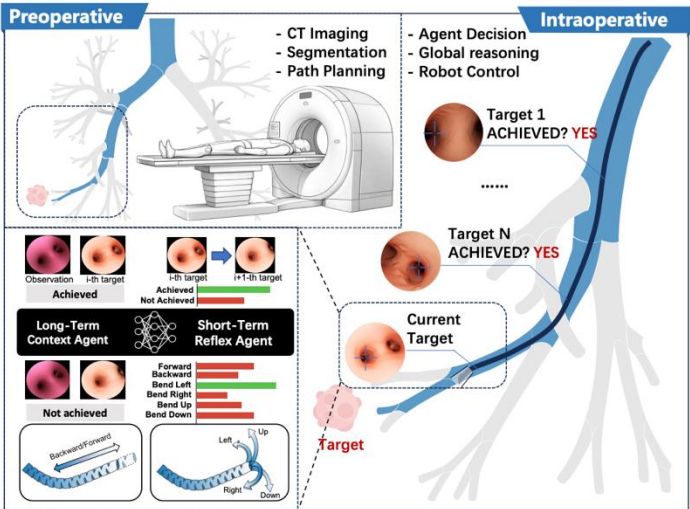
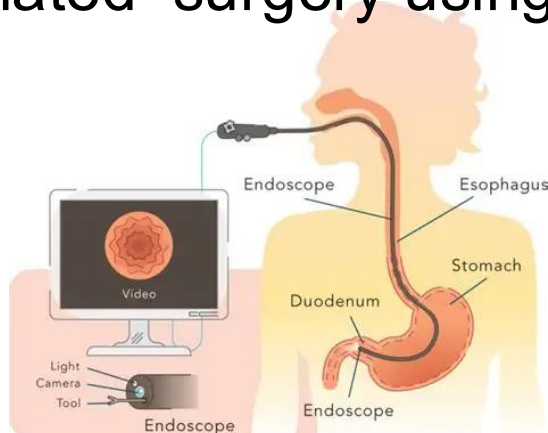
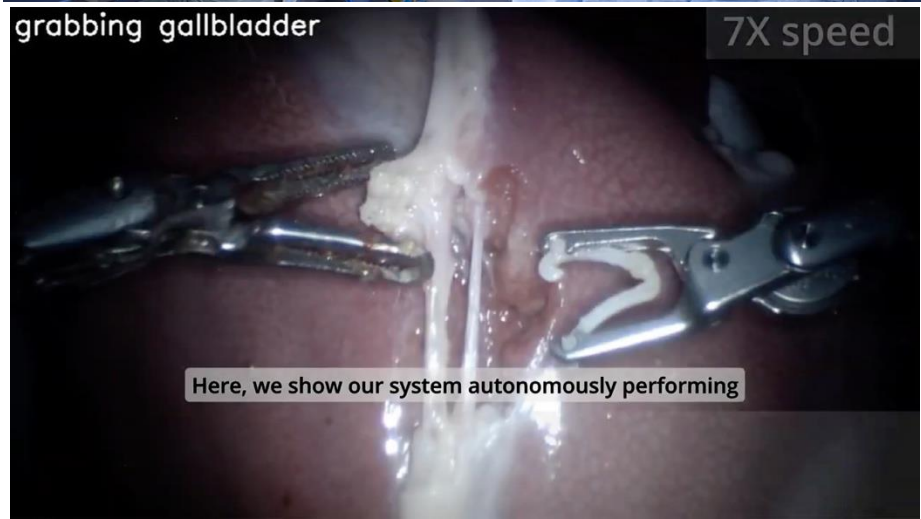
Mohism Lab's Research Focus and Direction towards the Goal

Research Focus: Machine Learning for Embodied Artificial Intelligence

Research Direction: ML for Intra-operative imaging for surgical robotics, medical imaging and diagnosis.

Automated surgery (da Vinci surgical robot)

Automated surgery using Notes



[1] Kim, Ji Woong, Juo-Tung Chen, Pascal Hansen, Lucy Xiaoyang Shi, Antony Goldenberg, Samuel Schmidgall, Paul Maria Scheikl et al. "SRT-H: A hierarchical framework for autonomous surgery via language-conditioned imitation learning." *Science robotics* 10, no. 104 (2025): eadt5254.
[2] Wu, Junyang, Mingyi Luo, Fangfang Xie, Minghui Zhang, Hanxiao Zhang, Chunxi Zhang, Junhao Wang, Jiayuan Sun, Yun Gu, and Guang-Zhong Yang. "Long-Short Term Agents for Pure-Vision Bronchoscopy Robotic Autonomy." *arXiv preprint arXiv:2603.07909* (2026).



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Embodied Robotics Team at Mohism Lab



Shiyuan Yang
PhD Candidate

Zhijia Tao
PhD Candidate

Borong Zhang
PhD Candidate

Xiaoming Liu
PhD Candidate
(joint Supervision)

Xiaofeng Kuang

Yilin Wang

UG: DUT PG:UM

UG: UESTC
PG: UM

PG: UM

UG: Sichuan
PG: UM

UG: SCAU(华农)
10 + Year work
experience, PG: UM

UG: UM
PG: UMN(US)

Focus: Multi-robot
Collaboration

Focus: Multi-robot
Coordination/Wheel-
legged Robot/Drone

Focus: VLA for
Robot Manipulation

Focus: Offline RL for
Vision based
Navigation

Focus: VLA for Robot
Manipulation

Summer
Research
Project Focus:
Multi-robot
Coordination



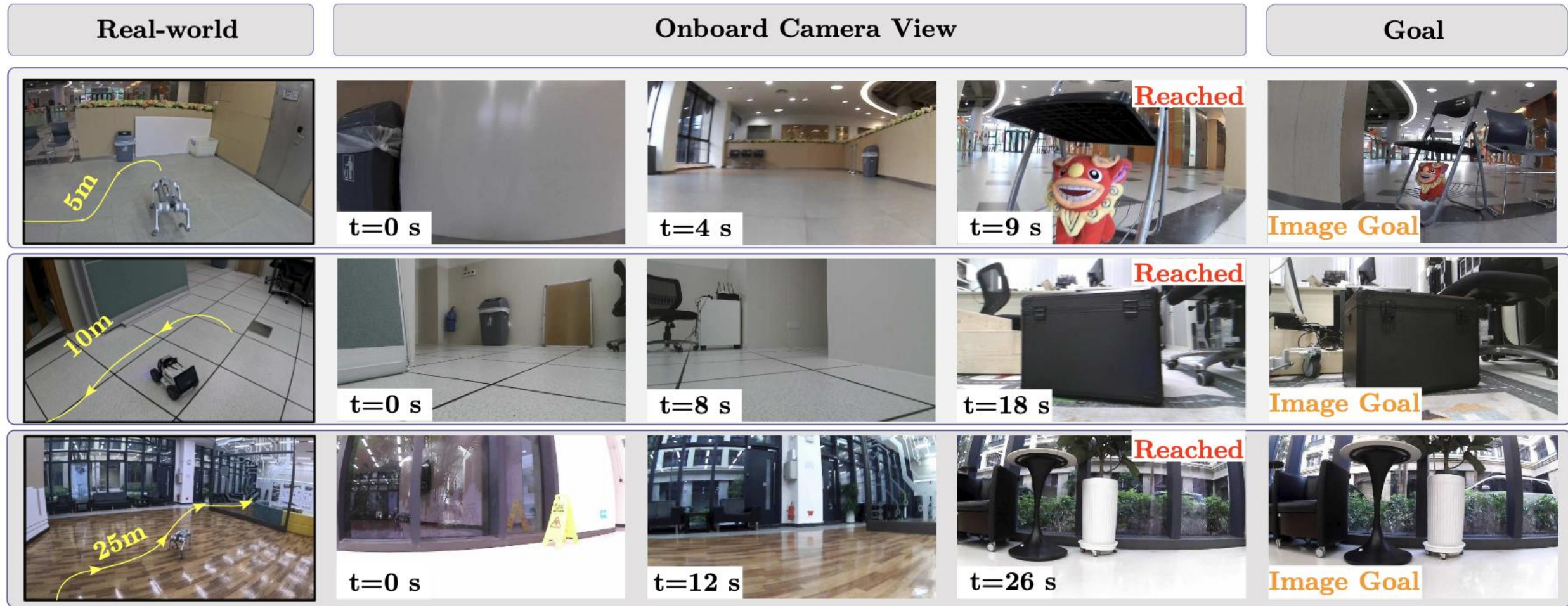
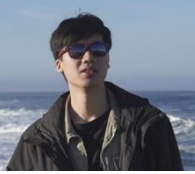
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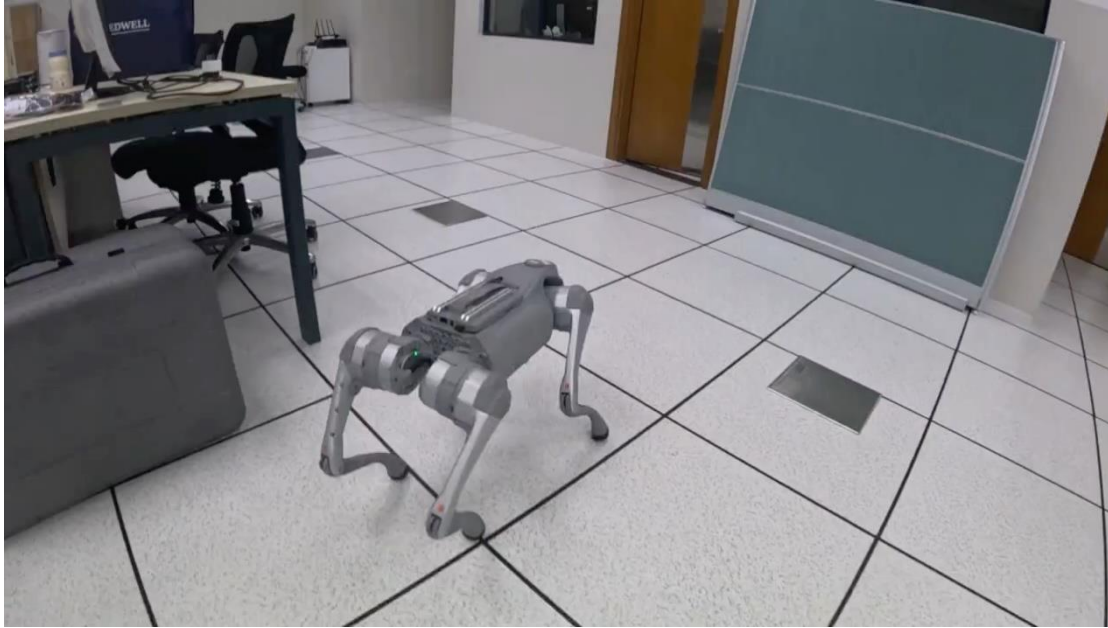
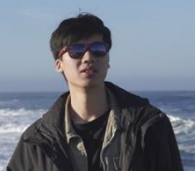
Single Agent Vision - 120 Minutes and a Laptop (AAAI 2027):

Minimalist Image-goal Navigation via Zero-shot Goal-conditioned Reinforcement Learning



Single Agent Vision - 120 Minutes and a Laptop (AAAI 2027):

Minimalist Image-goal Navigation via Zero-shot Goal-conditioned Reinforcement Learning



Robot View

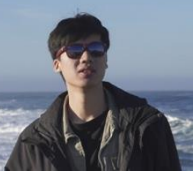
Goal Image



Robot View

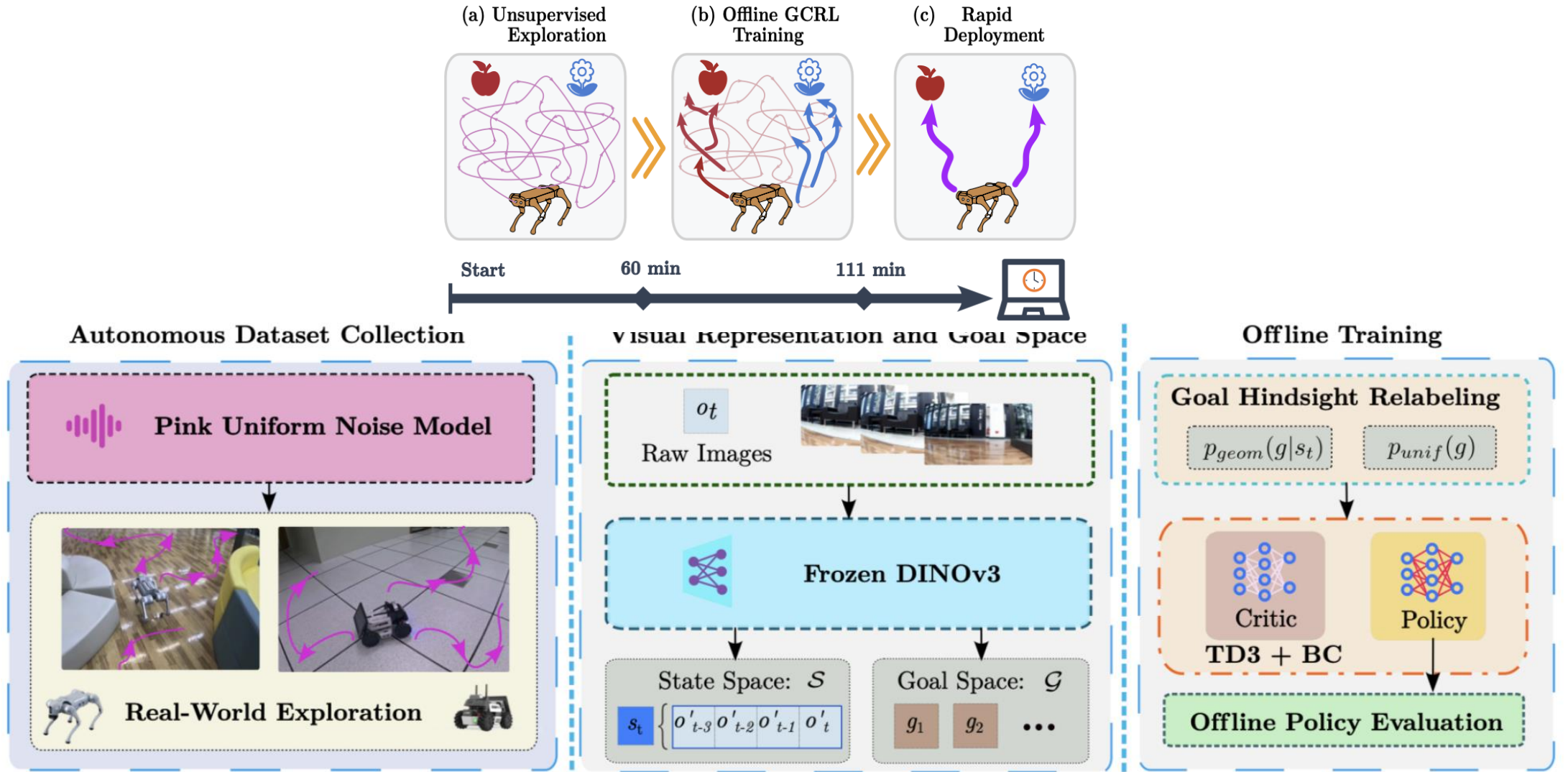
Goal Image



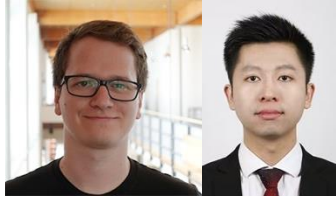


Single Agent Vision - 120 Minutes and a Laptop (AAAI 2027):

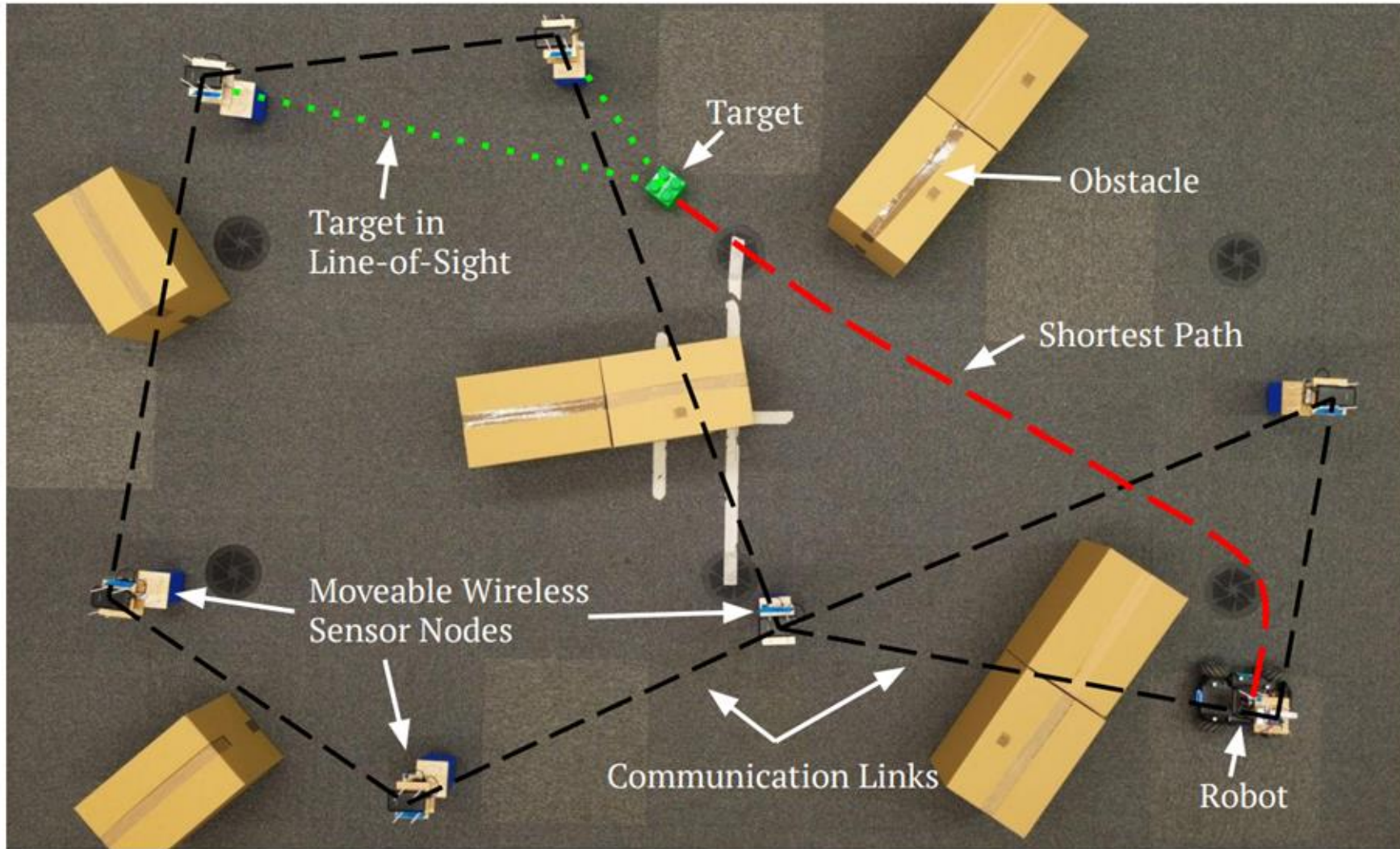
Minimalist Image-goal Navigation via Zero-shot Goal-conditioned Reinforcement Learning



Multi-agent Collaboration: Robot Navigation with Shared Vision



System design

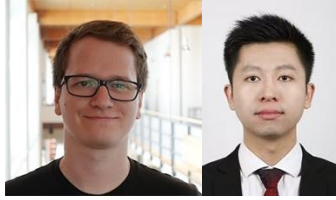


See What the Robot Can't See: Learning Cooperative Perception for Visual Navigation

Jan Blumenkamp, Qingbiao Li, Binyu Wang, Zhe Liu and Amanda Prorok



Multi-agent Collaboration: Robot Navigation with Shared Vision



Problem Definition

We formulate the navigation problem as a sequential decision-making problem at every time instant t

Each sensor S^I capture an omni-directional RGB image o_t^I

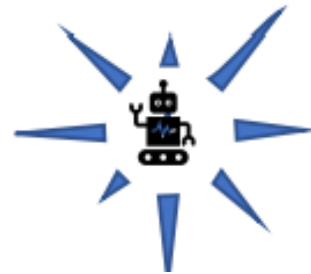
Each sensor predict the direction u_t^I out of eight along the shortest path towards the target G

Training paradigm:

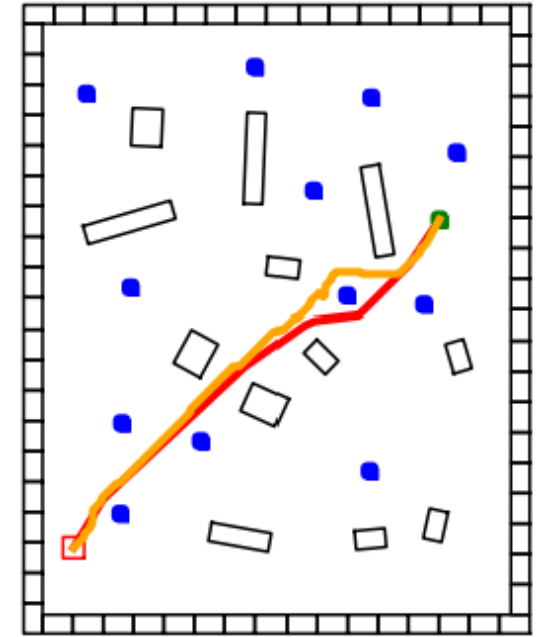
imitation learning (expert algorithm: Dijkstra's shortest path on visibility graphs)



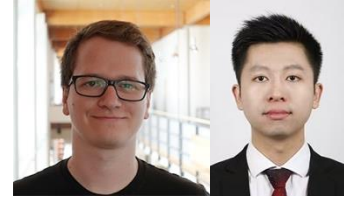
Observation o_t^i



Action



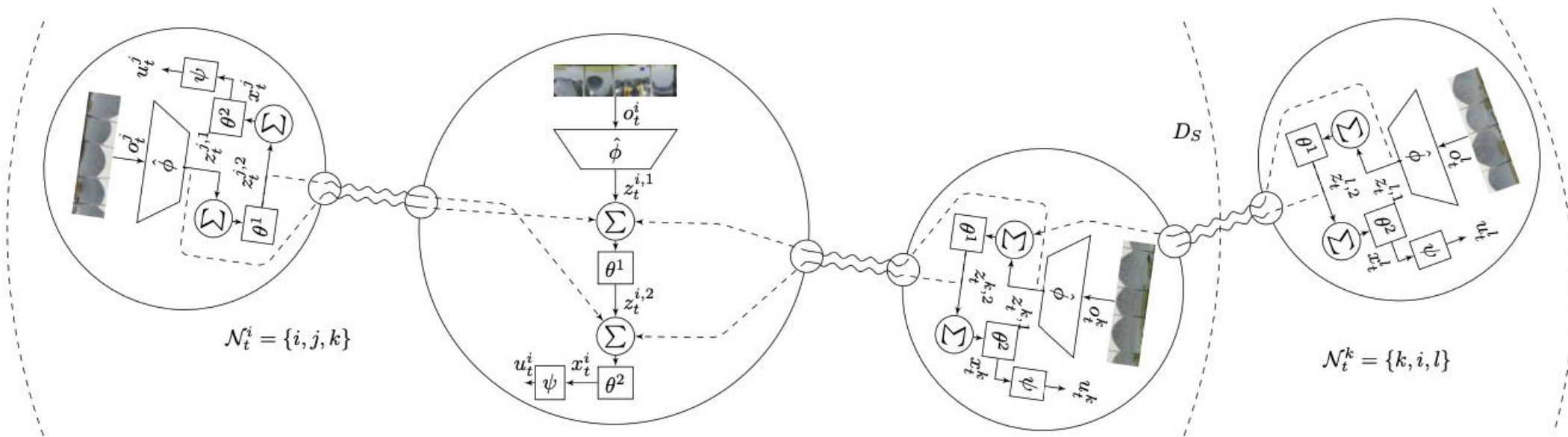
Multi-agent Collaboration: Robot Navigation with Shared Vision Framework



Local feature extraction: CNN – MobileNet v2

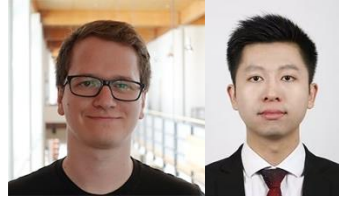
Neighborhood feature aggregation via GNN where graph connectivity (adjacency matrix is defined by distance)

Action prediction: Multi-layer Perceptron (MLP) to predict a set of cost-to-go advantage for sample a target direction from it.

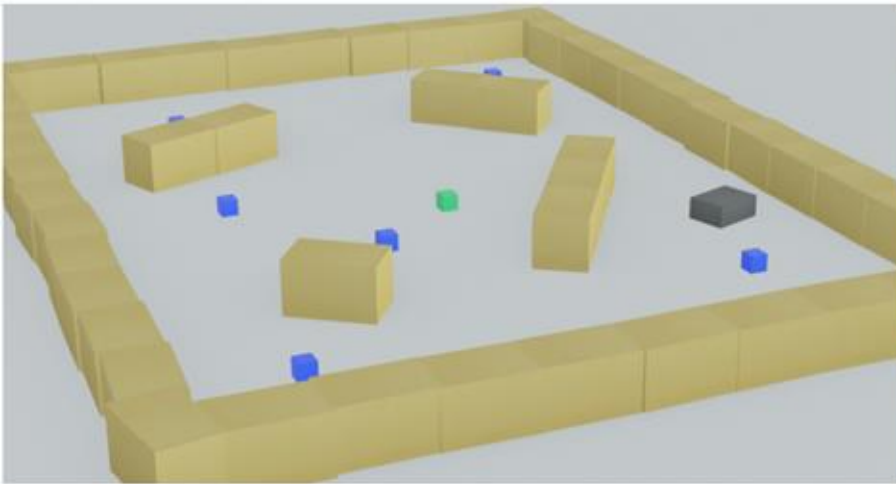


Multi-agent Collaboration: Robot Navigation with Shared Vision

System Design



Twin Environments



(b) Webots



(c) Real world



(d) Sensor



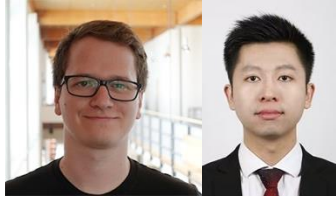
(e) Robot

See What the Robot Can't See: Learning Cooperative Perception for Visual Navigation

Jan Blumenkamp, Qingbiao Li, Binyu Wang, Zhe Liu and Amanda Prorok



Multi-agent Collaboration: Robot Navigation with Shared Vision



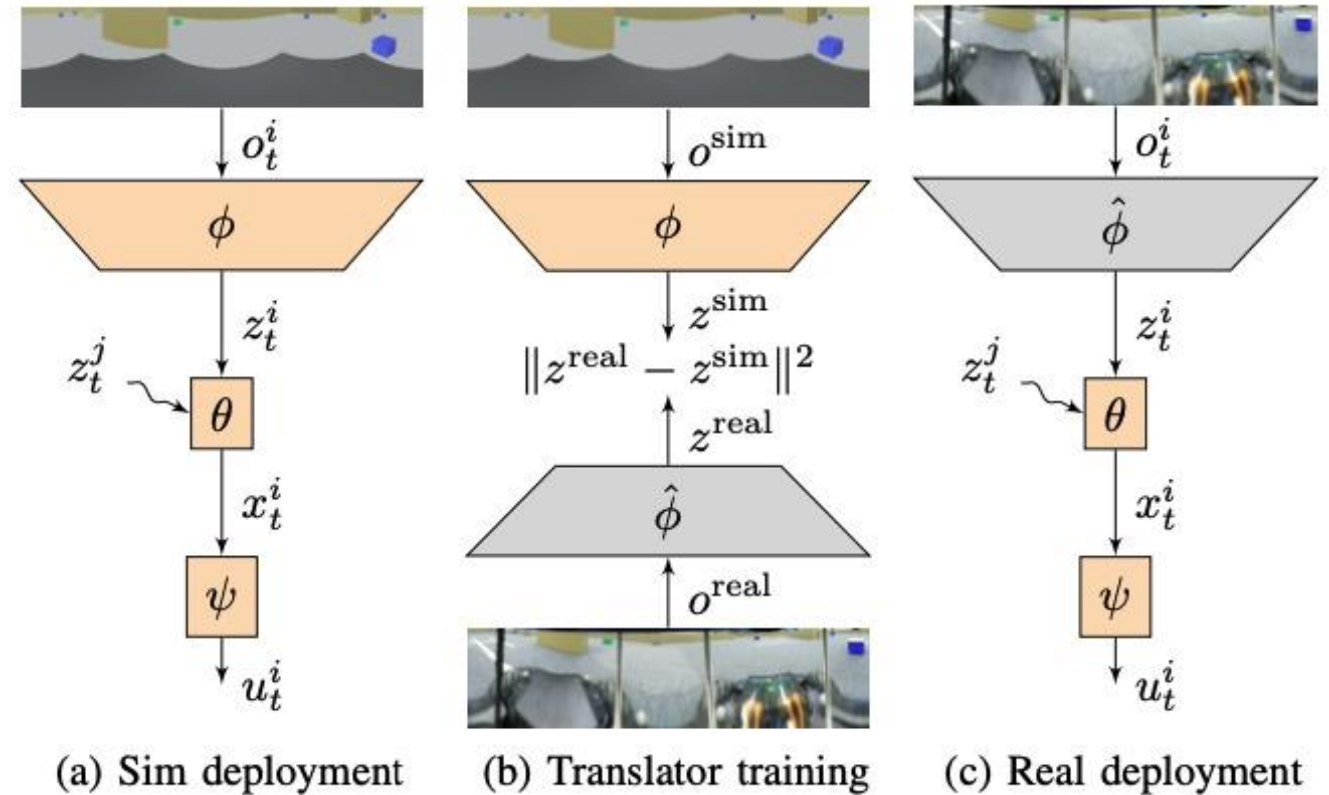
Domain Adaption

1. Collect **M** image pairs

- Remote control robot in real world and capture its position using motion capture system to sync the position of robot in simulation
- Real and simulated images from webots.

2. Use the image pair to train the translation model

3. Zero-shot transfer for real world deployment



See What the Robot Can't See: Learning Cooperative Perception for Visual Navigation

Jan Blumenkamp, Qingbiao Li, Binyu Wang, Zhe Liu and Amanda Prorok

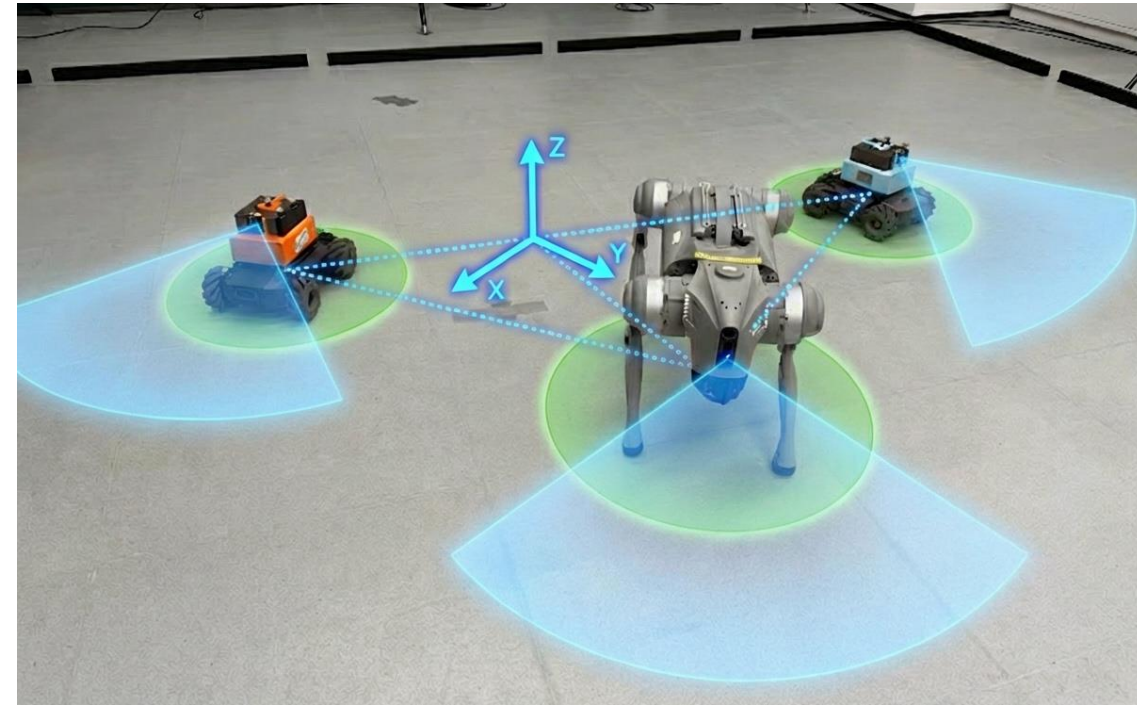
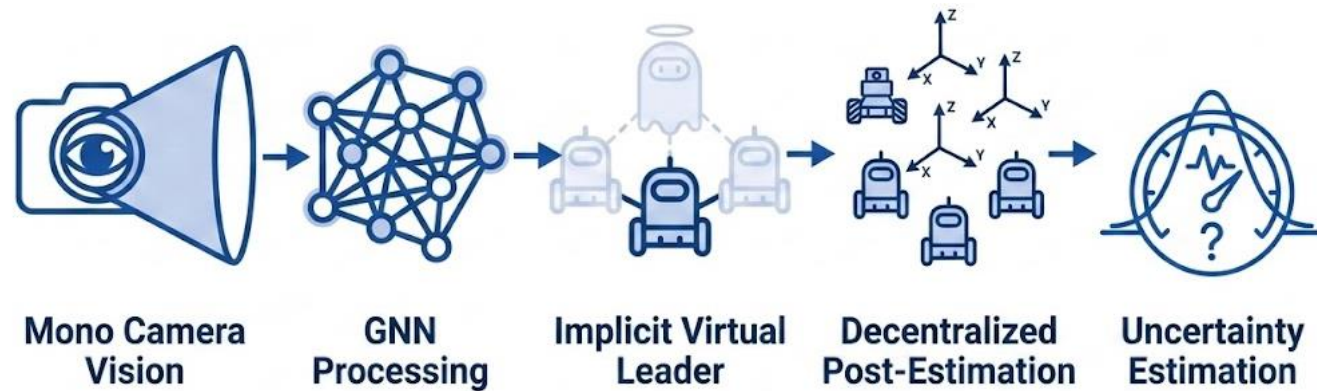




Multi-agent Collaboration: GNN IVL

Decentralized Vision-Only Formation Control via Graph Neural Networks with Uncertainty-Aware Relative Pose Estimation

We propose a novel end-to-end approach that eliminates the need for external dependencies. This method is based solely on airborne visual perception and point-to-point communication, achieving autonomous estimation of relative poses among robots and supporting decentralized formation maintenance.



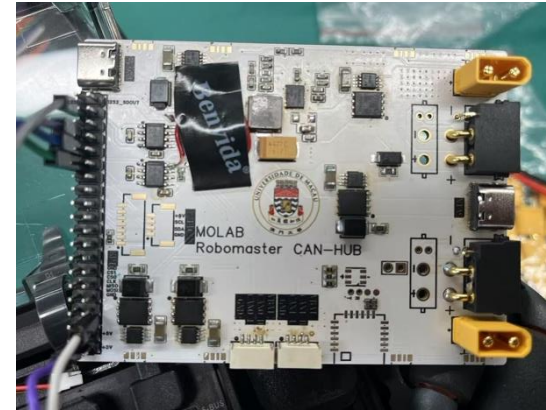
Multi-agent Collaboration: GNN IVL

Decentralized Vision-Only Formation Control via Graph Neural Networks with Uncertainty-Aware Relative Pose Estimation

Laboratory equipment



Jetson AGX Orin



Control PCB



Unitree Go2

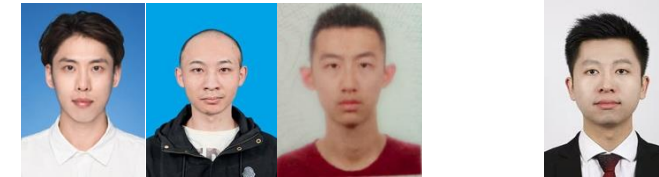


Mohism robomaster



**Implicit Virtual Leader: Decentralized Vision-Only Formation Control
via Graph Neural Networks with Uncertainty-Aware Relative Pose Estimation**

VLA for Robot Manipulation (Metaworld Rank 1st, AAAI 2027)



物智进化 EVOMIND Evo-Studio Embodied AI

9,802 24,337 中文 VLA DEX 登录

筛选 开源, SFT, RL 显示全部指标

标准评测模型 按平均分排名。点击行查看子指标。 17 个模型

排名 ↑	模型	分享	点赞	平均分	日期	论文	代码
1	FabriVLA 开源 SFT original	分享	1	90.0	2026-07	链接	代码
2	SUREFlow 开源 SFT original	分享	1	88.3	2026-07	链接	代码
3	LA4VLA-1B 开源 SFT original	分享	2	87.5	2026-06	链接	代码

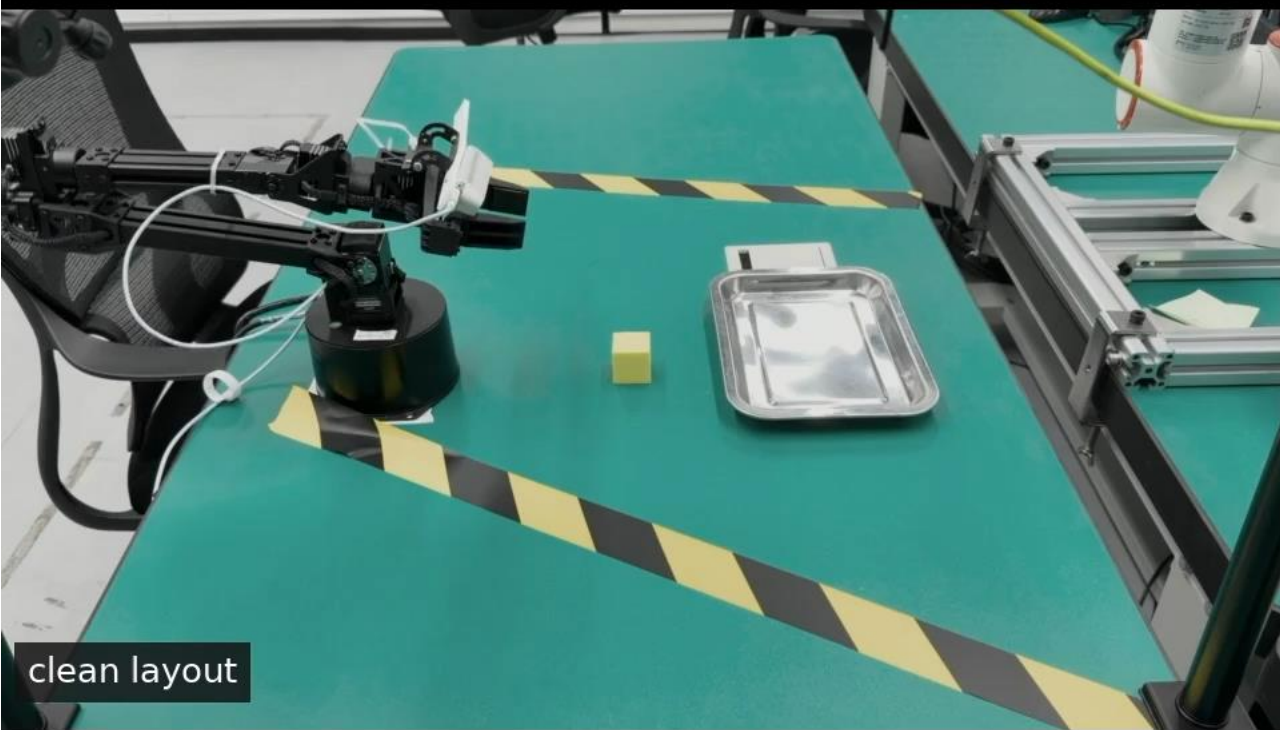
Shiyuan Yang, TaoZhijia, Borong Zhang, Jizheng Zhang, Junfei Guo, Yun Gu, Donglai Ran, Bian xu, Qingbiao Li, FabriVLA: A Lightweight Vision-Language-Action Model with Conformal Action Chunk Uncertainty, the Association for the Advancement of Artificial Intelligence (AAAI) (CCF A) (Under Review). 2026.



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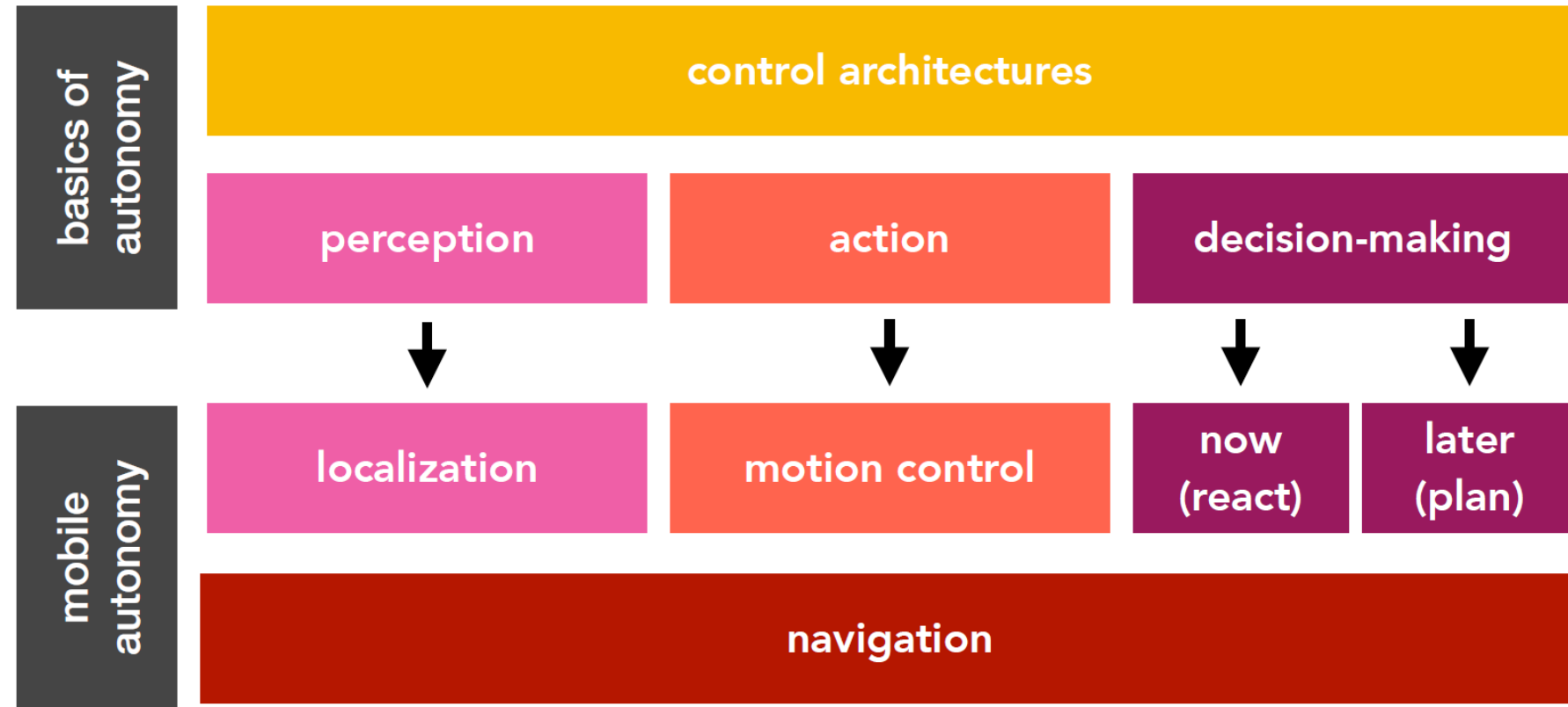
clean layout



objects beside the transfer path

What the Field of Robotics Studies?

- Learn foundational methods
 - Perception
 - Motion Control
 - Planning
 - Automation



Basics of Autonomy

Challenges:

How to model the perceive the world?

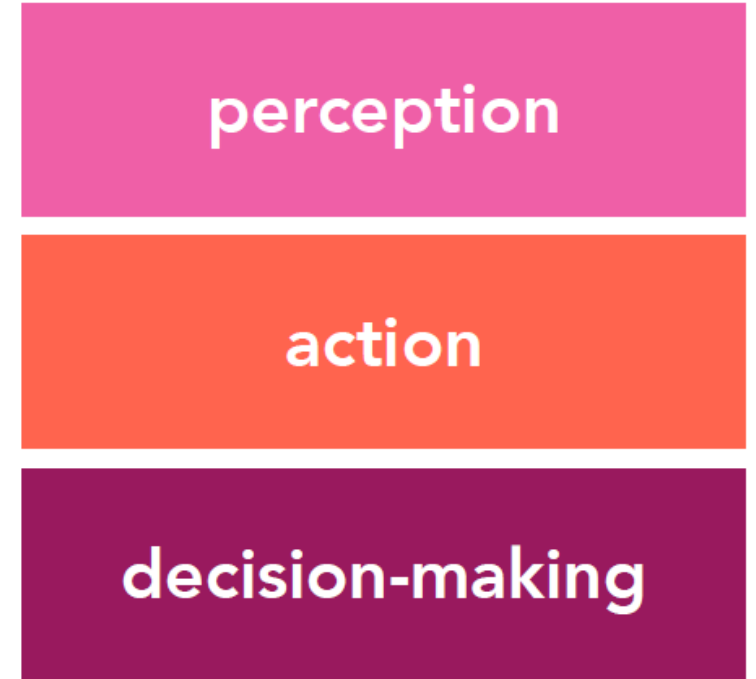
➤ Robot perception

How to process information and execute actions?

➤ Action

How to reason and plan in the face of uncertainty?

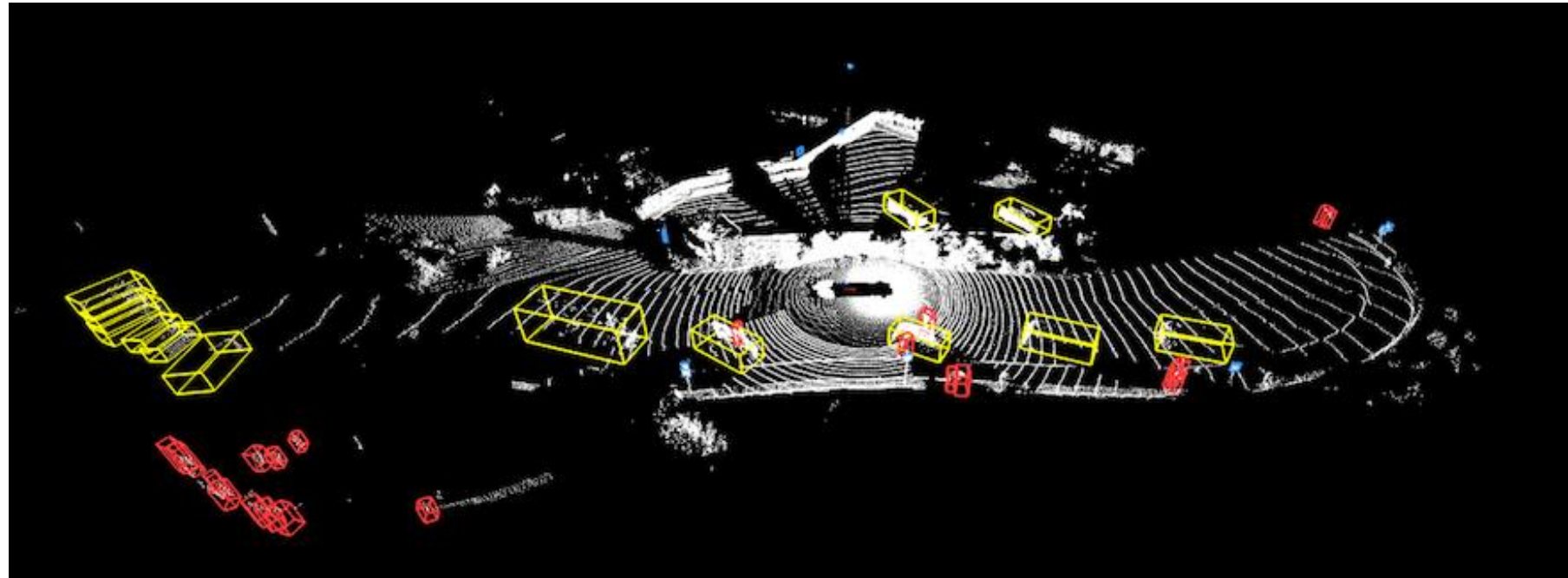
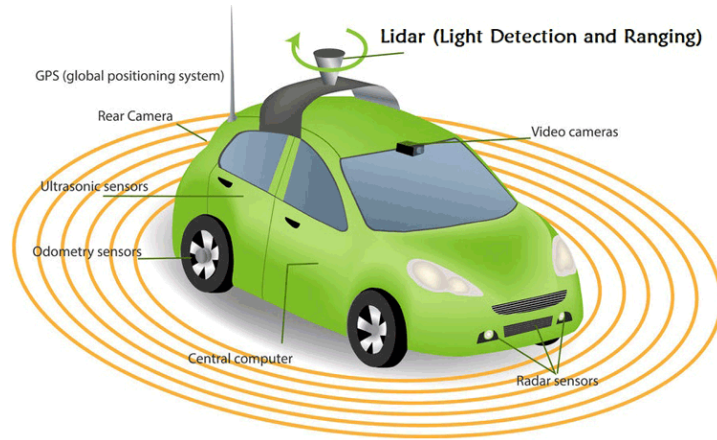
➤ Decision-Making:



Basics of Autonomy: Perception

Where am I? What am I doing?

Example: LIDAR



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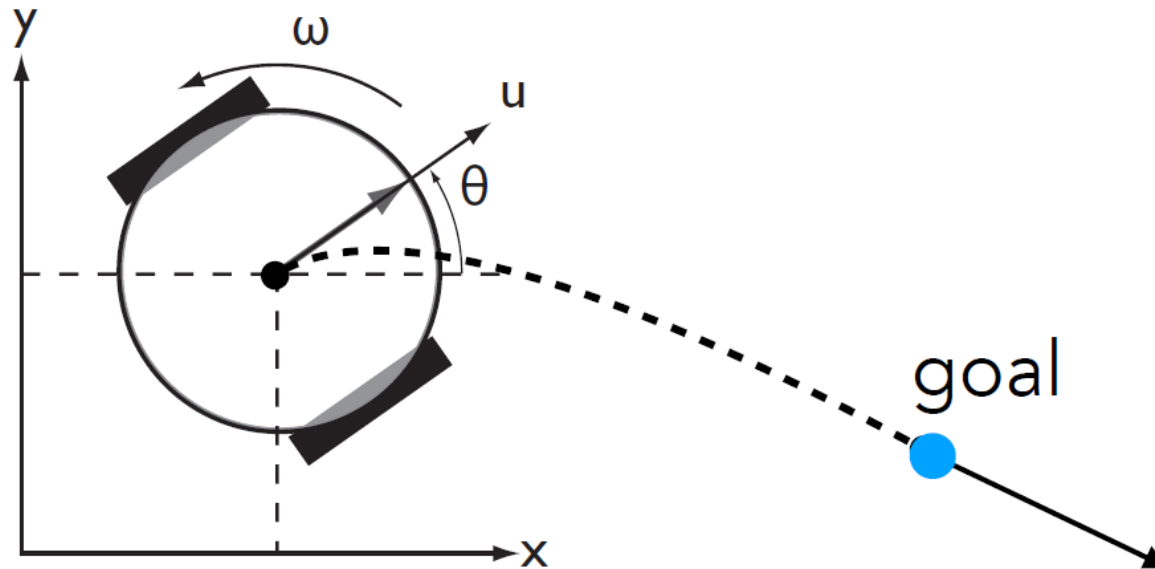
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Basics of Autonomy: Action

What force should my motors exert?

Example (motion control for a wheeled robot, PID controller):

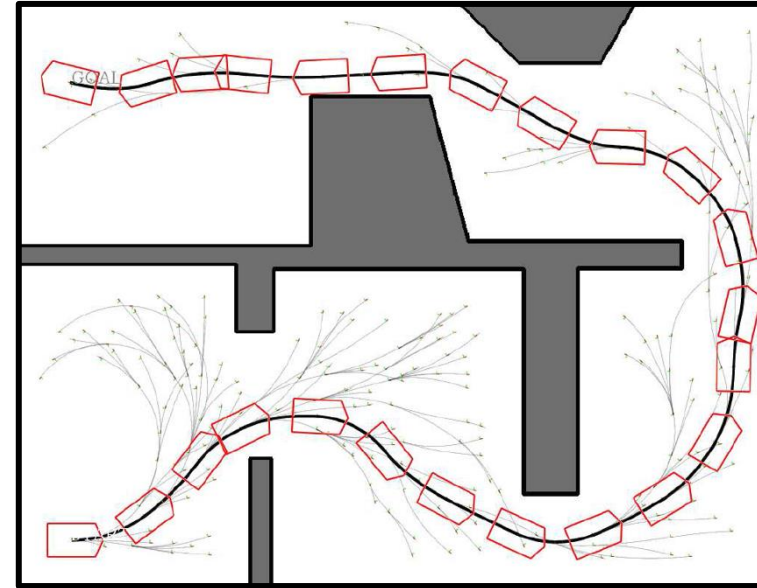
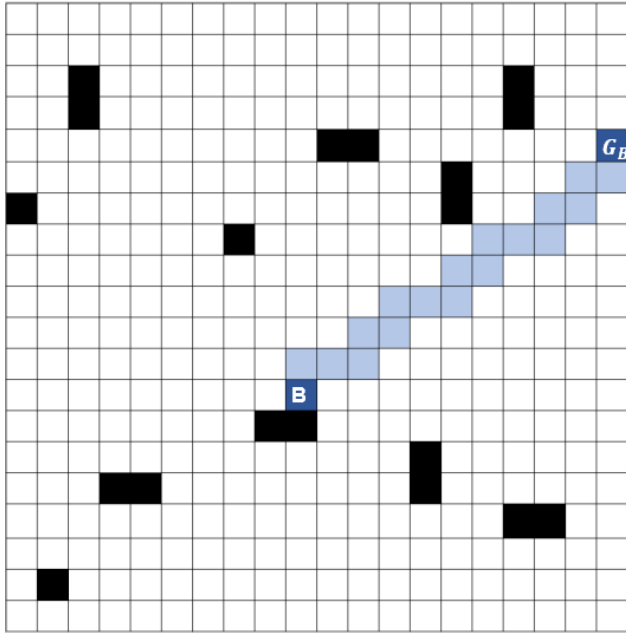
- Compute rotational and forwards velocities (or acceleration).



Basics of Autonomy: Decision-Making

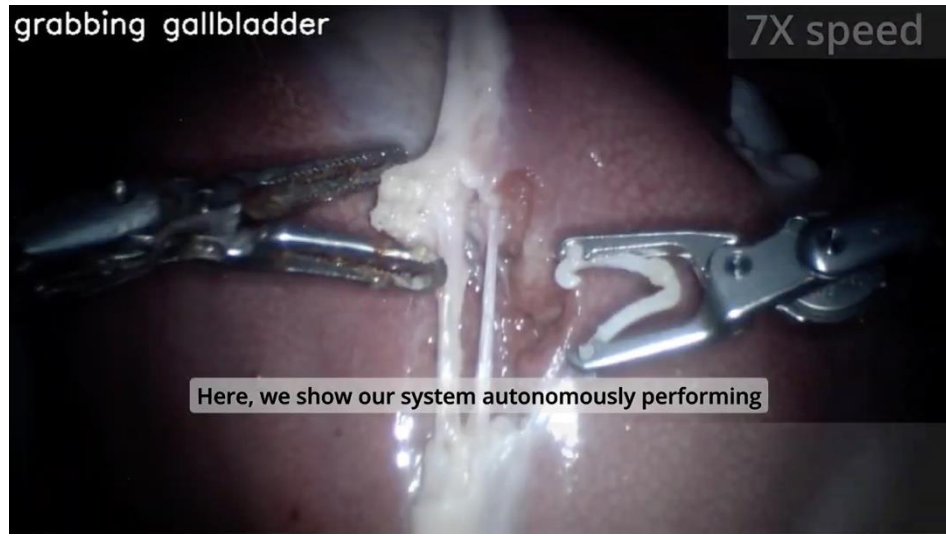
What is my plan to achieve my goal?

- Example: (deliberative planning, A* or RRT*)
- Compute a path that does not collide with obstacles and that respects the robot's motion constraints.

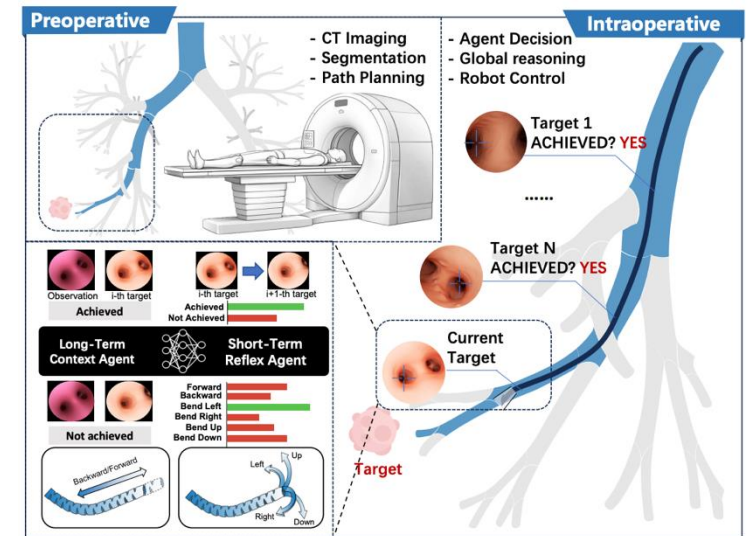
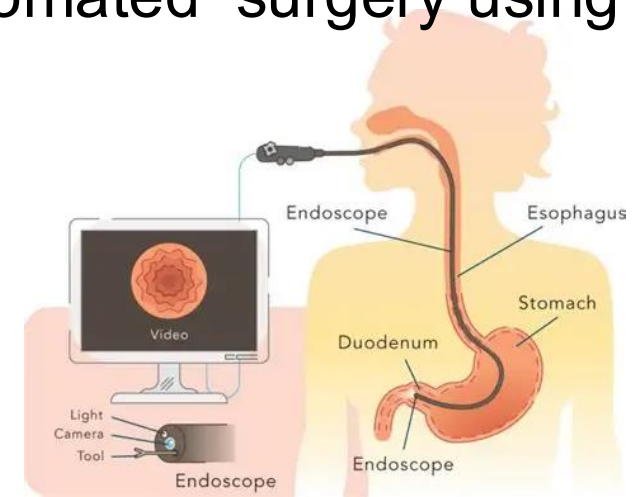


The field of Surgical Robotics and Today's Focus

Automated surgery (da Vinci surgical robot)



Automated surgery using Notes



[1] Kim, Ji Woong, Juo-Tung Chen, Pascal Hansen, Lucy Xiaoyang Shi, Antony Goldenberg, Samuel Schmidgall, Paul Maria Scheikl et al. "SRT-H: A hierarchical framework for autonomous surgery via language-conditioned imitation learning." *Science robotics* 10, no. 104 (2025): ead15254.
[2] Wu, Junyang, Mingyi Luo, Fangfang Xie, Minghui Zhang, Hanxiao Zhang, Chunxi Zhang, Junhao Wang, Jiayuan Sun, Yun Gu, and Guang-Zhong Yang. "Long-Short Term Agents for Pure-Vision Bronchoscopy Robotic Autonomy." *arXiv preprint arXiv:2603.07909* (2026).

Surgical Robotics Team at Mohism Lab and Collaborators



Qin Liu
PhD Candidate

PG: Sydney

Focus: Semantic
Segmentation
model for
Surgical
Robotics



Huanrong Liu
PhD Candidate

UG: China
University of
Geosciences
PG: UM

Focus: World
Action model for
Surgical
Robotics



Yanchen MU
PhD Candidate
(UM-PolyU joint
Program)

UG: Xi'an
Jiaotong
University
PG: Peking U

Focus: Surgical
video generation
model, Surgical
world model



Keru Zheng

UG: CUHK(SZ)

Focus: Multi-
Agent
Frameworks for
Medical Image
Analysis and
Diagnosis



Zhouxuan Xia
Research
Assistant
(Remote)

PG: Imperial
College London

Focus: Surgical
Scene
Understanding
Surgical
Robotics



Tianci Yang
Research
Assistant

UG: Beijing
Normal
University
Zhuhai

Focus: Build up
semi-auto
labelling system
and framework
system



Kejian Shi
Collaborator
from CUHK

PhD: CUHK at
Prof Dou

Focus: Human-
in-Loop for Safe
Surgical Robot
Control



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Perception for Intelligent Surgical Robotics: SIA-RAPN

SIA-RAPN (Suturing Recognition and Analysis in Robot Assisted Partial Nephrectomy Dataset)



Action recognition

4128段缝合手势



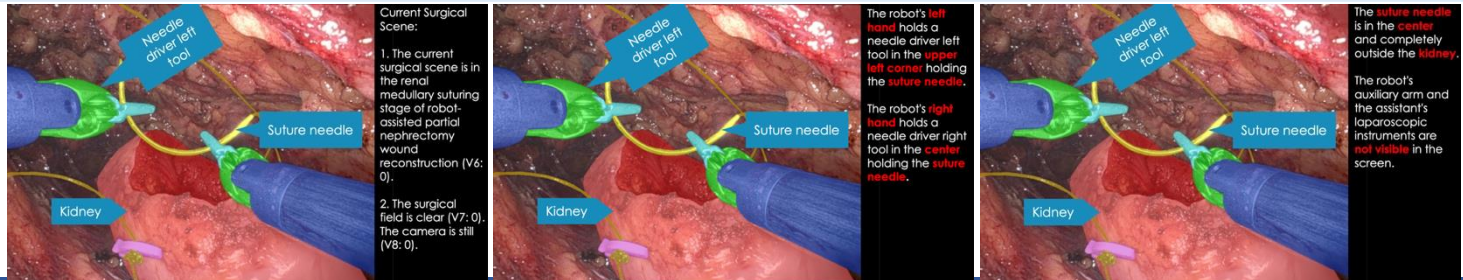
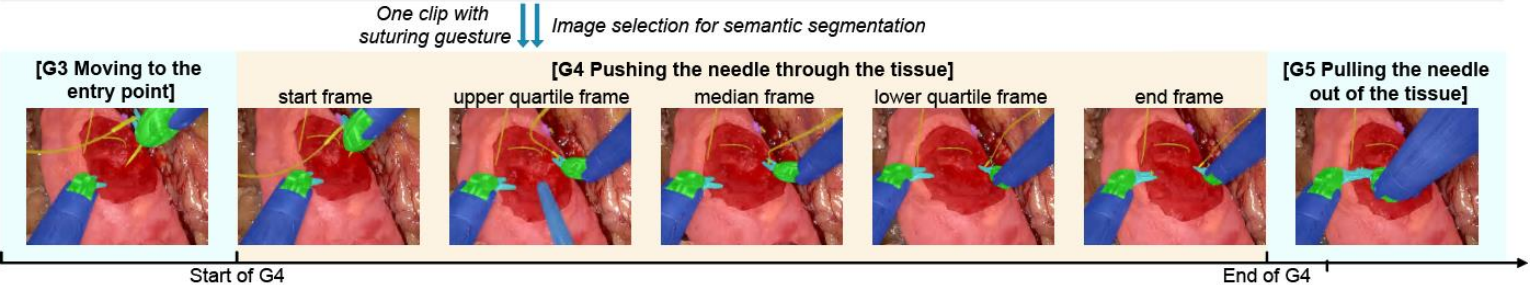
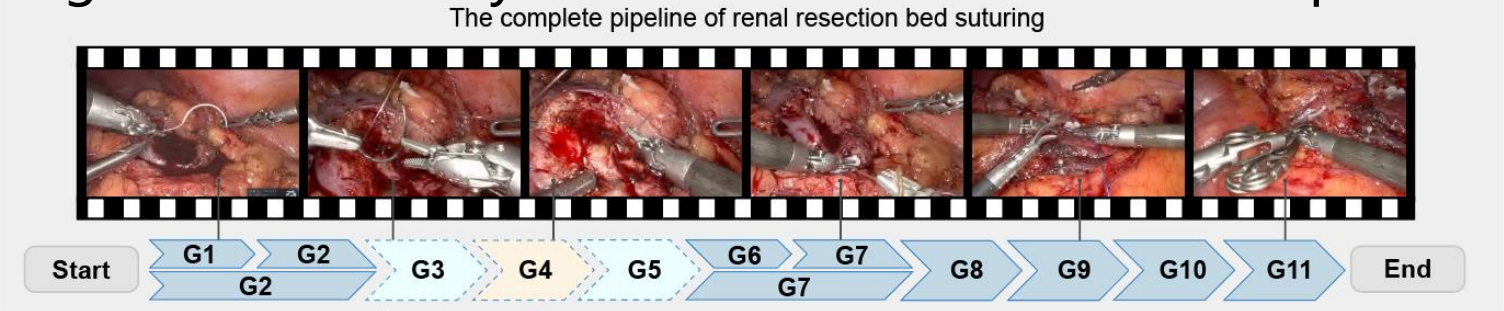
Semantic Segmentation

20640张图像分割



Surgical Strategy

1531个缝合策略



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Perception for Intelligent Surgical Robotics: SIA-RAPN Action Recognition

Gesture index	Gesture description
G1	Reaching for needle with right hand
G2	Positioning needle
G3	Pushing needle through tissue
G4	Transferring needle from left to right
G5	Moving to center with needle in grip
G6	Pulling suture with left hand
G7	Pulling suture with right hand
G8	Orienting needle
G9	Using right hand to help tighten suture
G10	Loosening more suture
G11	Dropping suture at end and moving to end points
G12	Reaching for needle with left hand
G13	Making C loop around right hand
G14	Reaching for suture with right hand
G15	Pulling suture with both hands.

JIGSAWS, 2014

ID	Gesture description
G0	Other
G1	Picking-up the needle
G2	Positioning the needle tip
G3	Pushing the needle through the tissue
G4	Pulling the needle out of the tissue
G5	Tying a knot
G6	Cutting the suture
G7	Returning/dropping the needle

IEEE TRANSACTIONS ON MEDICAL IMAGING, 2022
MICAI Challenge, 2023



ID	肾脏创面缝合手势
G0	其他
G1	持针
G2	调针
G3	运针
G4	进针
G5	出针
G6	理线
G7	上hem-o-lock夹
G8	打结
G9	剪线
G10	放置止血绵
G11	释放Bulldog夹
G12	收针

G2 调针

G3 运针

G4 进针

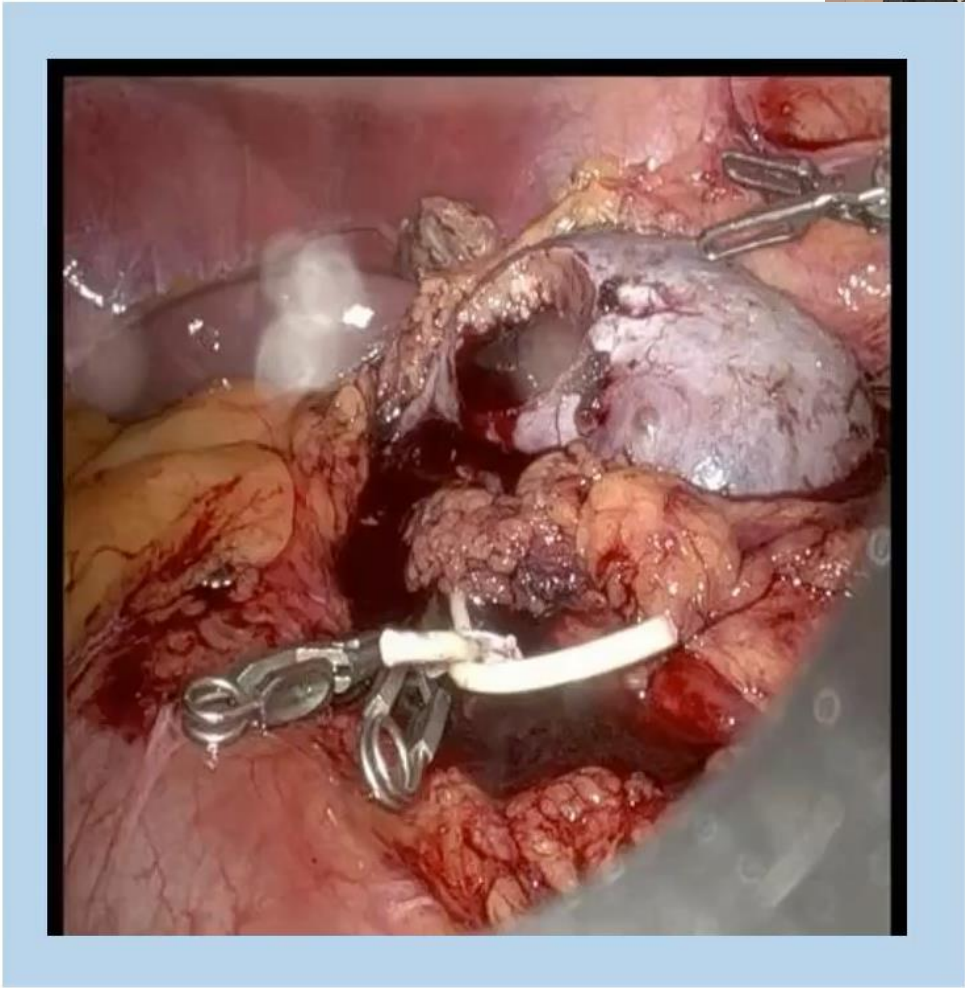
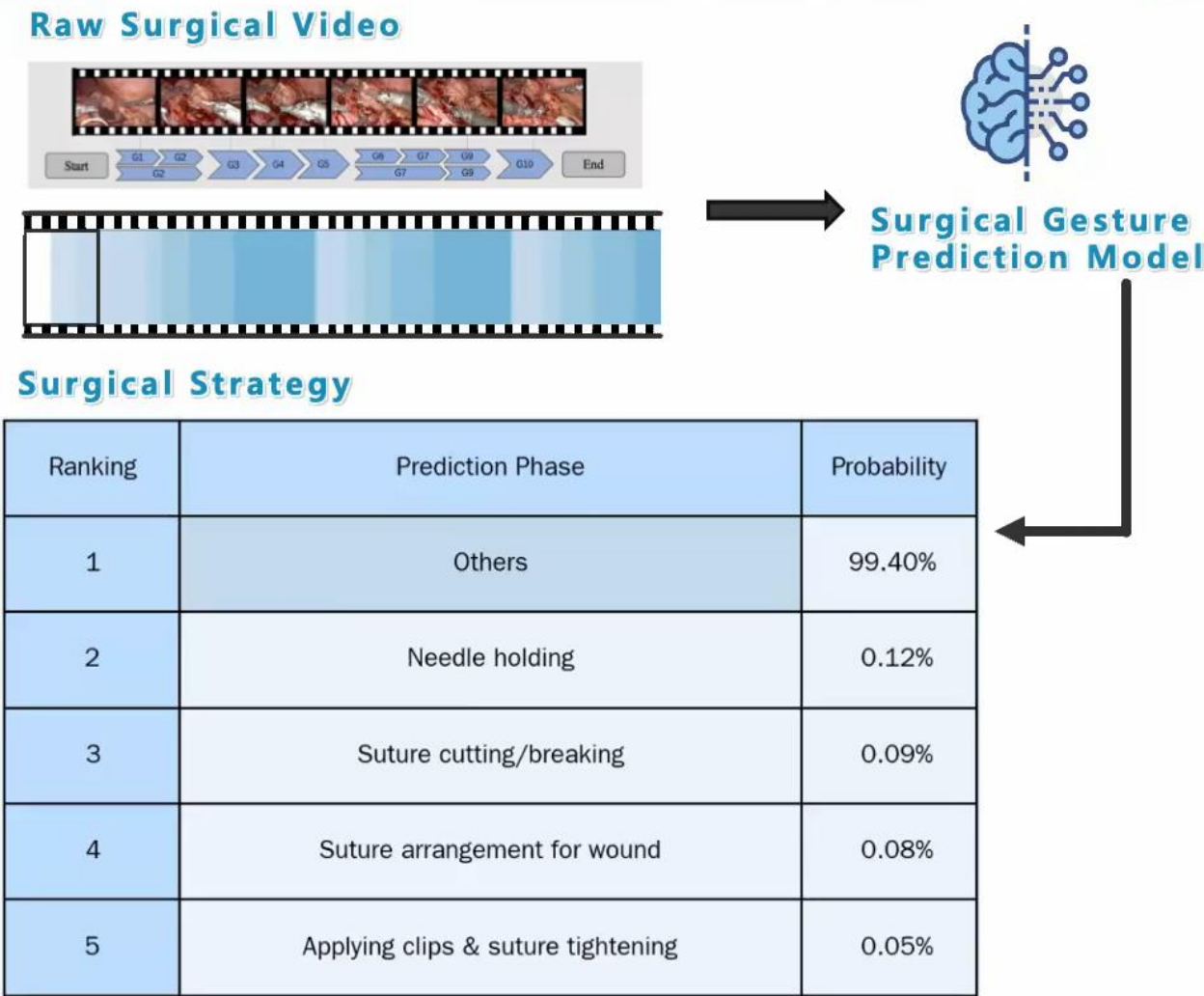
G5 出针



Perception for Intelligent Surgical Robotics: SIA-RAPN

Action Recognition

Surgical Gesture Prediction Model on Raw Surgical Video (Our Result Visualization)



Perception for Intelligent Surgical Robotics: SIA-RAPN

Action Recognition



Model	F1@10	F1@25	F1@50	Acc	Bal-Acc	Edit	Frame mAP
MS-TCN++	76.54	71.96	58.02	68.83	67.54	74.93	48.79
AsFormer	79.60	73.78	57.87	64.55	58.73	78.00	40.08
TUT	76.04	71.03	57.62	61.86	54.57	76.28	37.61
DiffAct	83.14	80.23	66.86	72.12	67.16	82.30	51.28

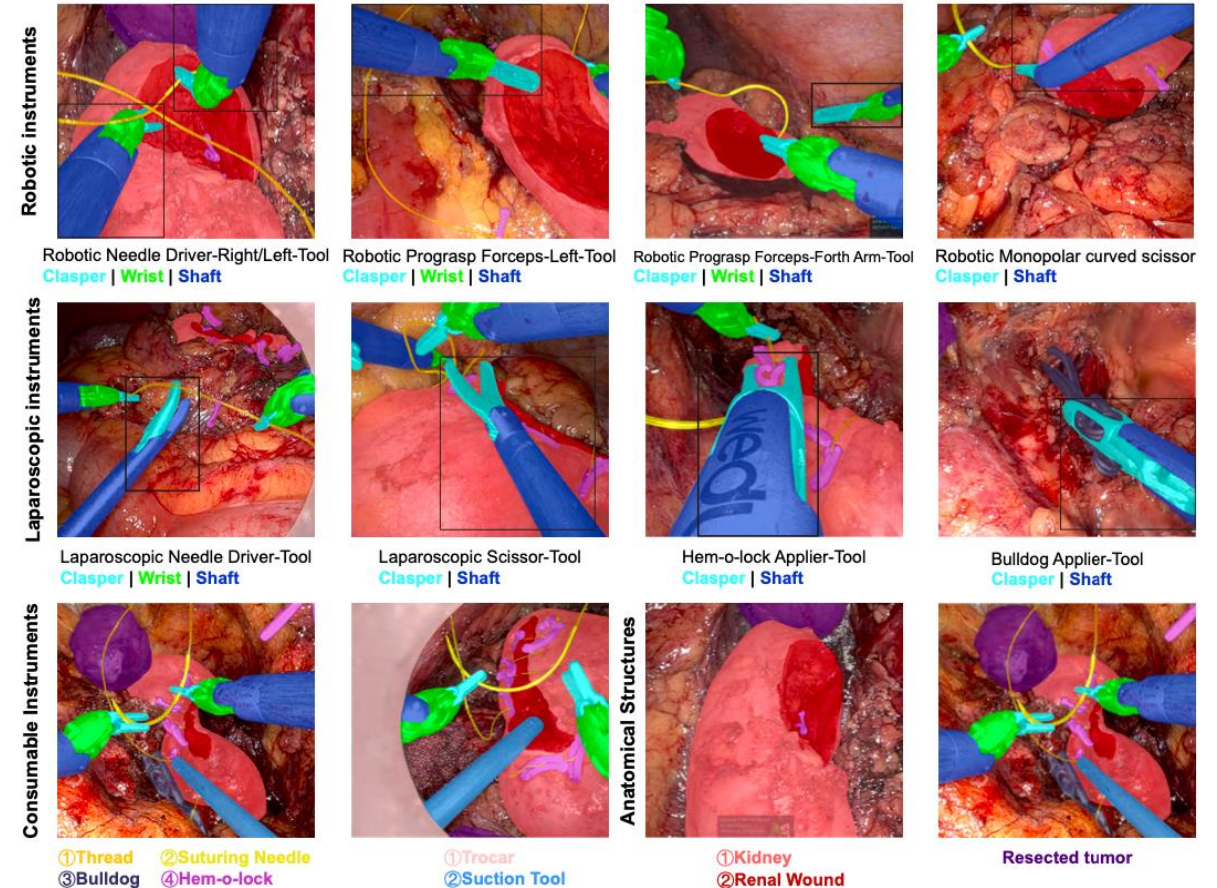
Perception for Intelligent Surgical Robotics: SIA-RAPN

Semantic Segmentation

Collaborated with Doctors to label 20640 images



■ Resection Bed margin ■ Teared capsule margin

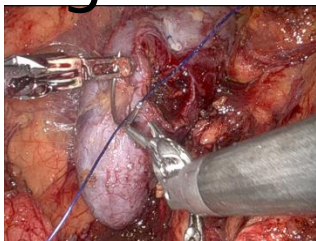


Perception for Intelligent Surgical Robotics: SIA-RAPN

Semantic Segmentation

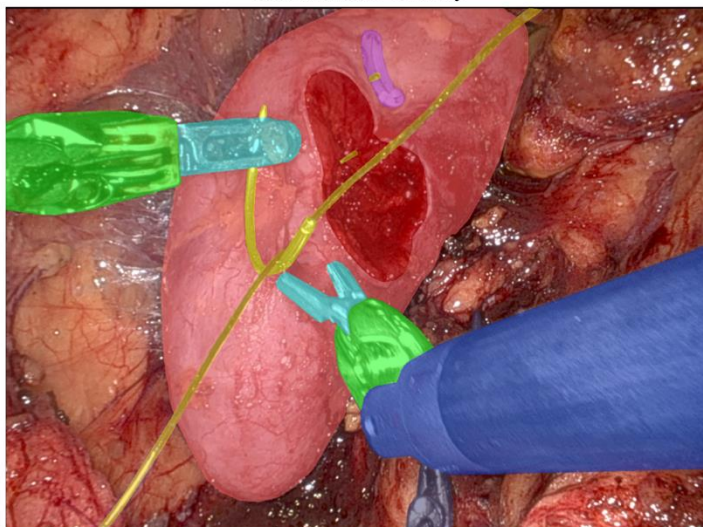


Surgical Video

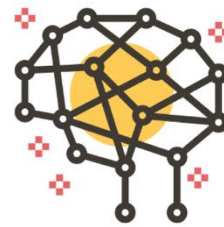


Expert Label

Ground Truth Overlay



- | | |
|--|-------------|
| Robotic Needle Driver-Right-Tool Clasper | Thread |
| Robotic Needle Driver-Right-Tool Wrist | Bulldog |
| Robotic Needle Driver-Right-Tool Shaft | Hem-o-lock |
| Robotic Prograsp Forceps-Tool Clasper | Renal Wound |
| Robotic Prograsp Forceps-Tool Wrist | Kidney |
| Suturing Needle | |



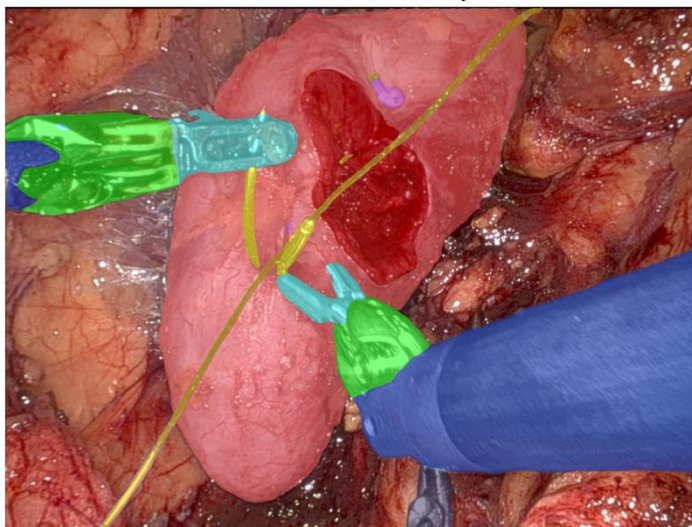
nnUNet



Prediction Accuracy

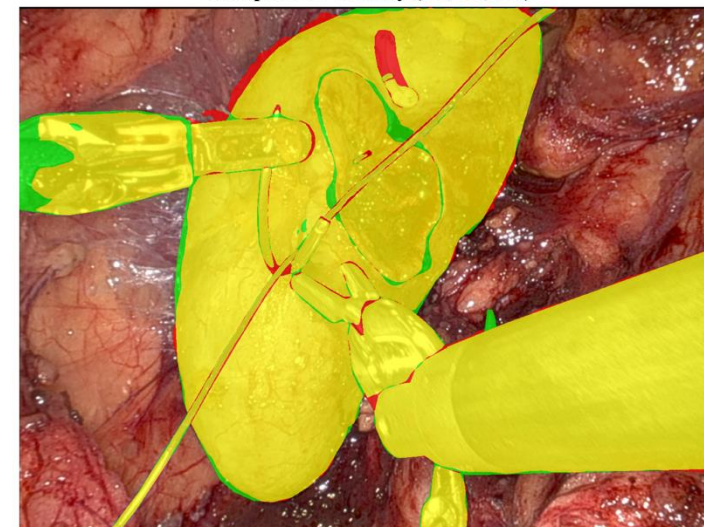
$mIOU = 0.81$, $mDICE = 0.89$
 $mHD95 = 131.86$
 $mPrecision = 0.94$, $mRecall = 0.86$

Predicted Mask Overlay



- | | |
|--|-----------------|
| Robotic Needle Driver-Right-Tool Clasper | Suturing Needle |
| Robotic Needle Driver-Right-Tool Wrist | Thread |
| Robotic Needle Driver-Right-Tool Shaft | Bulldog |
| Robotic Prograsp Forceps-Tool Clasper | Hem-o-lock |
| Robotic Prograsp Forceps-Tool Wrist | Renal Wound |
| Robotic Prograsp Forceps-Tool Shaft | Kidney |
| Suction Tool | |

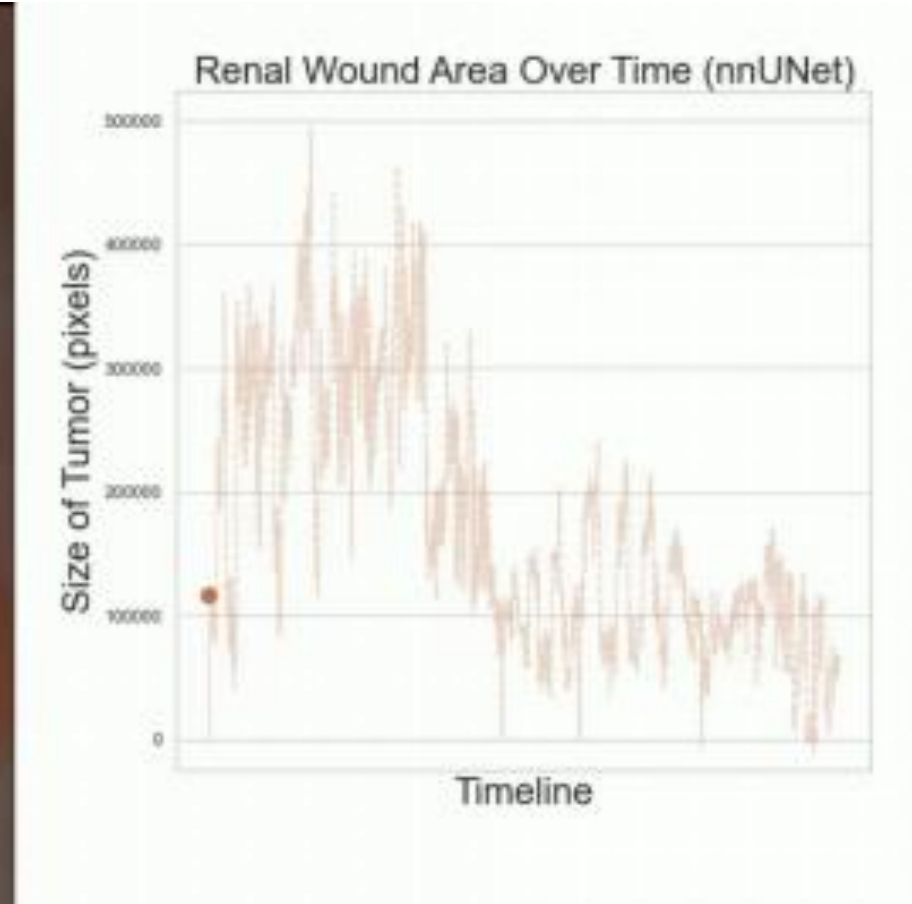
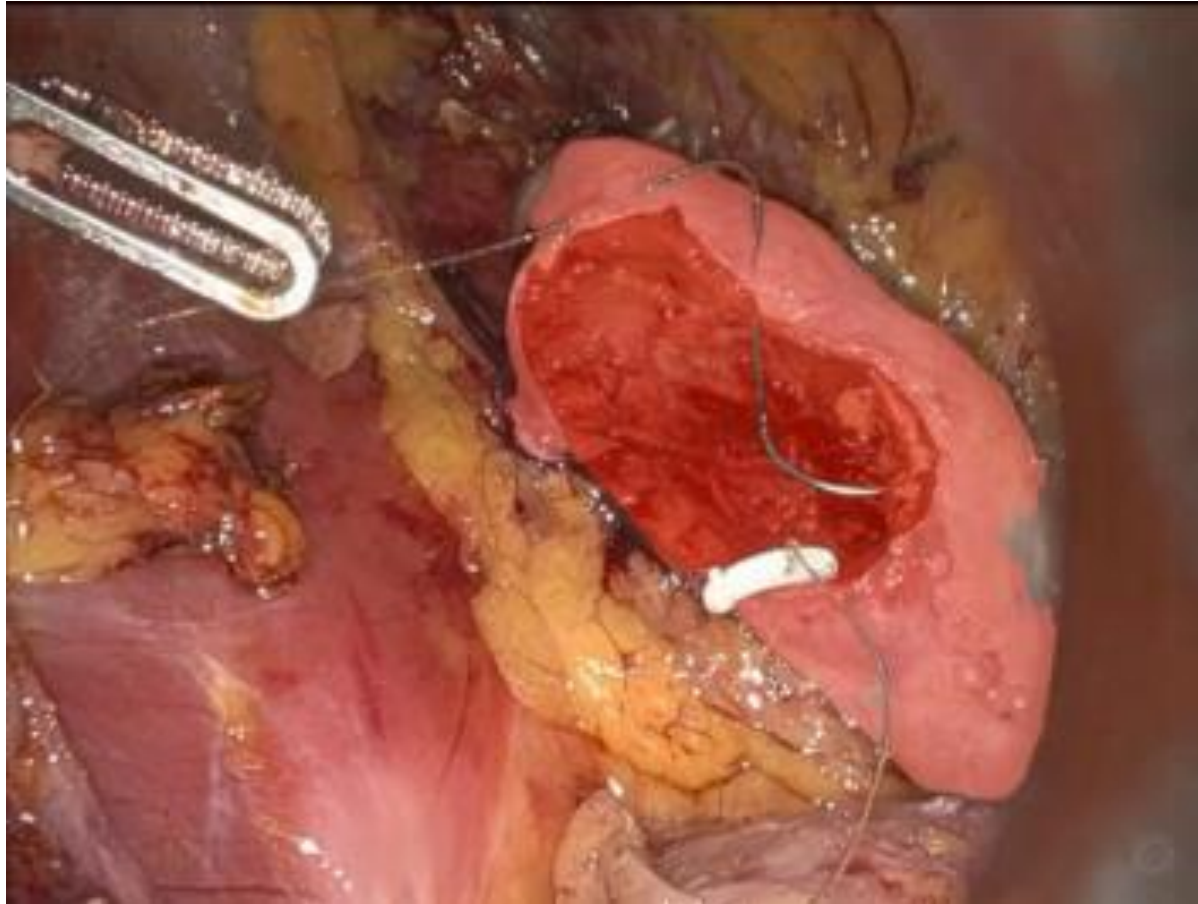
Comparison Overlay (TP, FP, FN)



- | | |
|--|----------------|
| | True Positive |
| | False Positive |
| | False Negative |

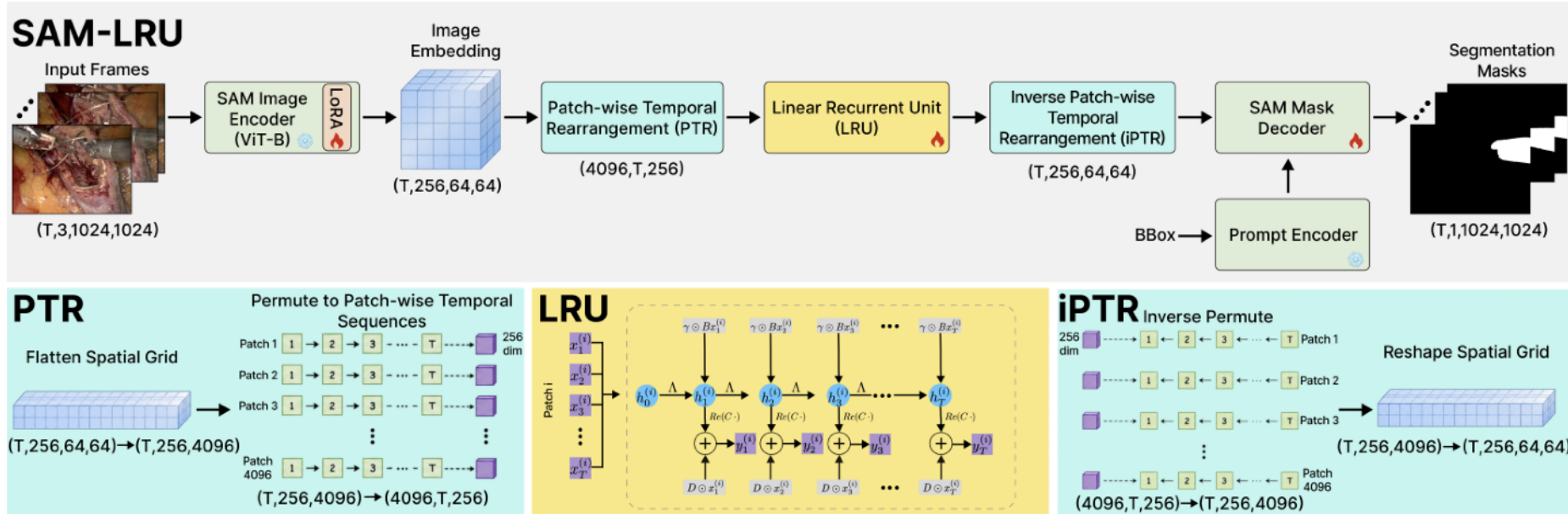
Perception for Intelligent Surgical Robotics: SIA-RAPN

Semantic Segmentation



Perception for Intelligent Surgical Robotics: SIA-RAPN

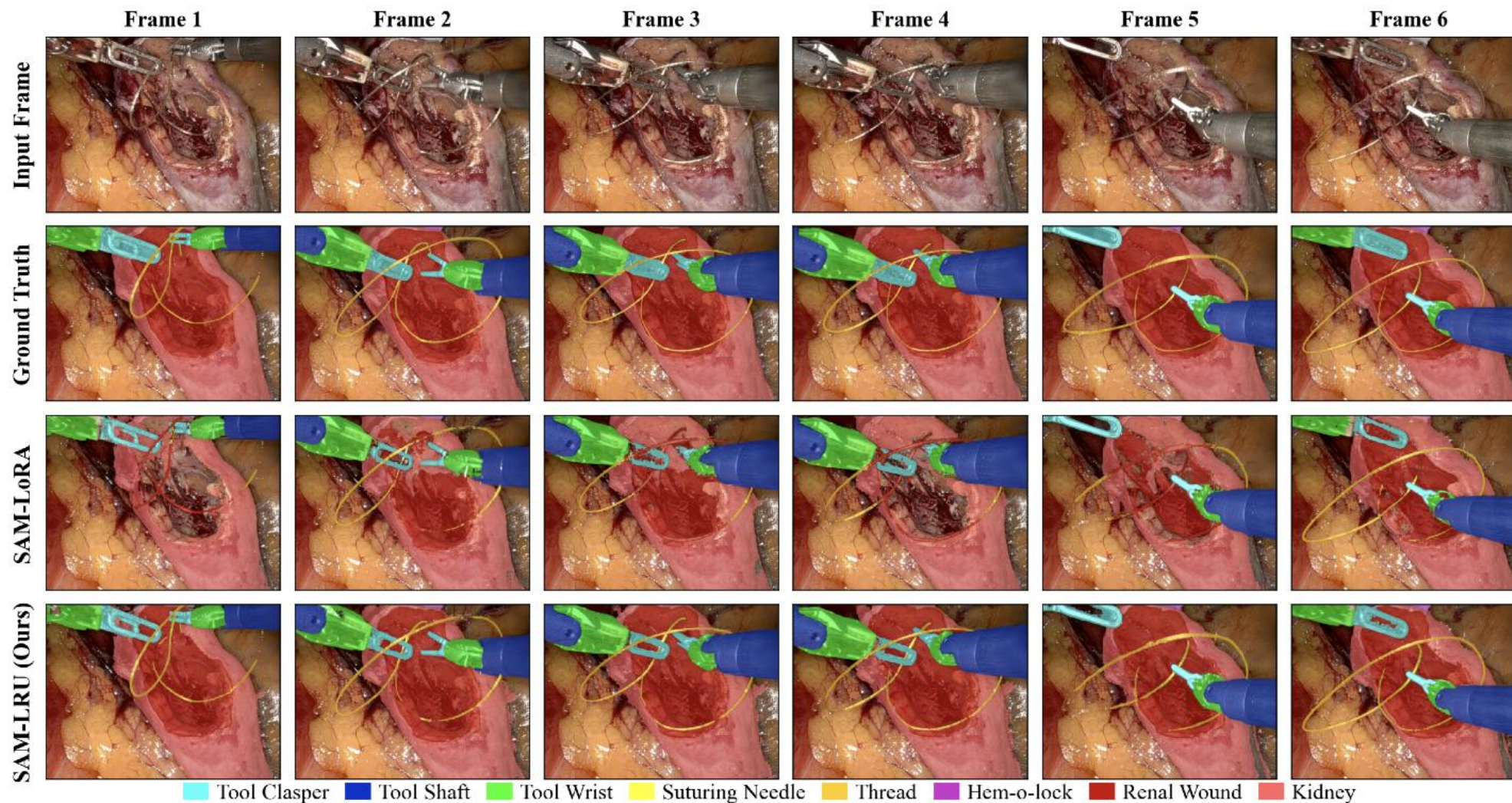
Semantic Segmentation



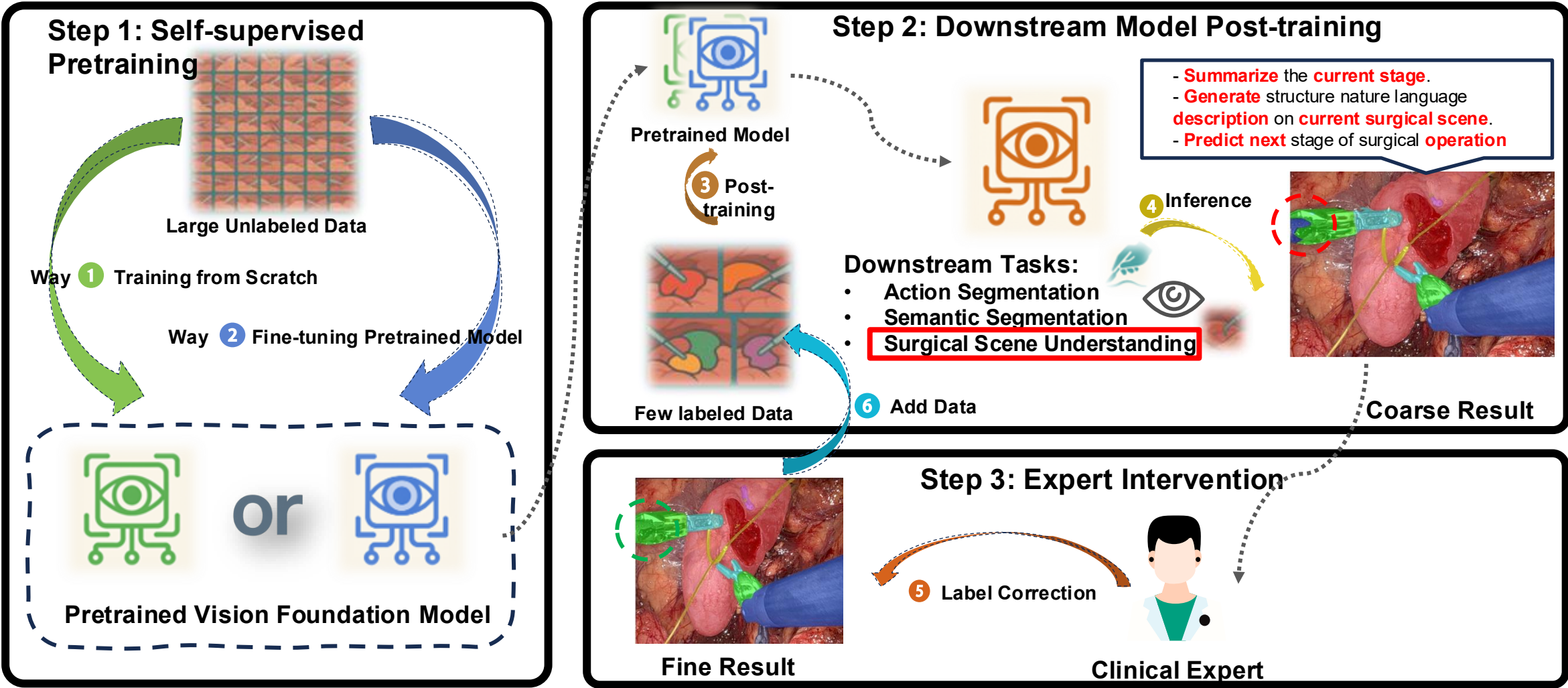
Method	Overall			Organ			Instrument		
	IoU $\uparrow$	Dice $\uparrow$	HD95 $\downarrow$	IoU $\uparrow$	Dice $\uparrow$	HD95 $\downarrow$	IoU $\uparrow$	Dice $\uparrow$	HD95 $\downarrow$
mU-Net [6]	0.701 $\pm$ 0.007	0.780 $\pm$ 0.006	103.85 $\pm$ 6.88	0.743 $\pm$ 0.016	0.812 $\pm$ 0.015	79.59 $\pm$ 2.68	0.690 $\pm$ 0.008	0.771 $\pm$ 0.008	110.46 $\pm$ 9.46
LongiSeg [16]	0.604 $\pm$ 0.011	0.688 $\pm$ 0.011	192.80 $\pm$ 9.22	0.747 $\pm$ 0.011	0.819 $\pm$ 0.011	92.56 $\pm$ 4.97	0.565 $\pm$ 0.014	0.652 $\pm$ 0.013	221.58 $\pm$ 11.99
SegFormer [19]	0.800 $\pm$ 0.001	0.866 $\pm$ 0.001	35.93 $\pm$ 0.81	0.812 $\pm$ 0.006	0.866 $\pm$ 0.007	40.14$\pm$2.75	0.796 $\pm$ 0.002	0.865 $\pm$ 0.001	34.82 $\pm$ 1.15
SAM-Med2D [2]	0.335 $\pm$ 0.020	0.460 $\pm$ 0.022	205.97 $\pm$ 16.56	0.321 $\pm$ 0.016	0.462 $\pm$ 0.018	226.36 $\pm$ 20.18	0.339 $\pm$ 0.024	0.460 $\pm$ 0.025	200.35 $\pm$ 17.29
SAM-LoRA [5]	0.793 $\pm$ 0.002	0.865 $\pm$ 0.001	28.59 $\pm$ 0.54	0.803 $\pm$ 0.008	0.873 $\pm$ 0.006	49.26 $\pm$ 2.87	0.791 $\pm$ 0.001	0.863 $\pm$ 0.001	22.94 $\pm$ 0.42
SAM-LRU (Ours)	0.812$\pm$0.001	0.885$\pm$0.001	24.11$\pm$0.36	0.816$\pm$0.004	0.885$\pm$0.004	45.67$\pm$1.19	0.811$\pm$0.001	0.884$\pm$0.001	18.23$\pm$0.17

Perception for Intelligent Surgical Robotics: SIA-RAPN

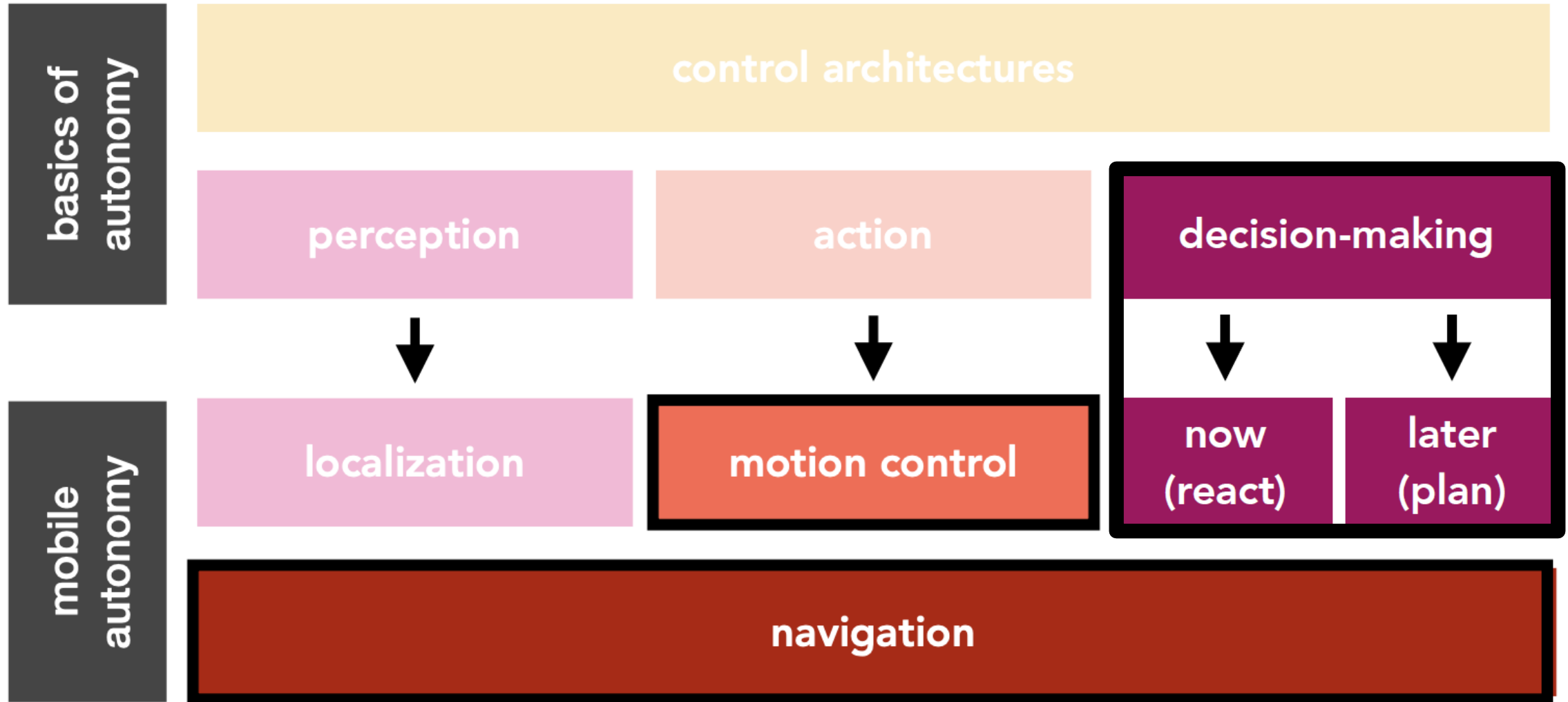
Semantic Segmentation



Perception for Intelligent Surgical Robotics: SIA-RAPN

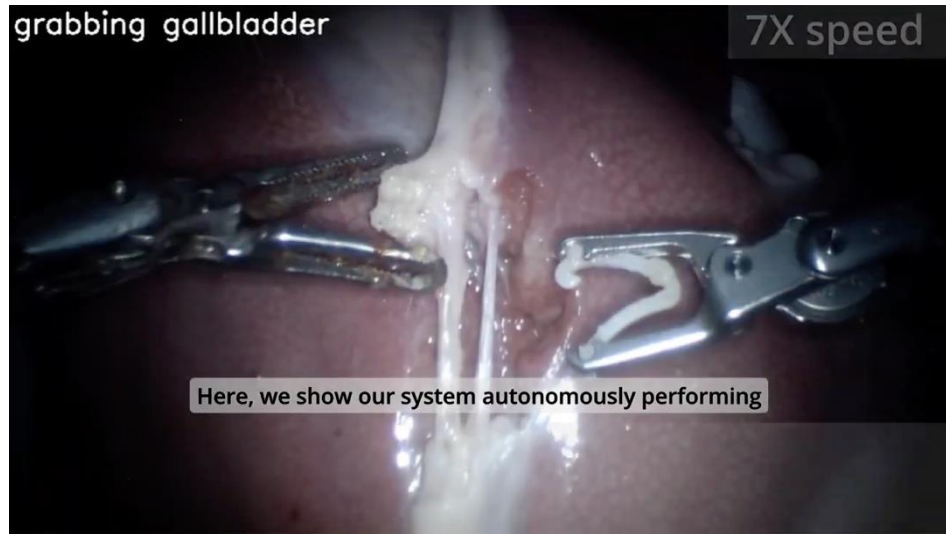


Basics of Autonomy: Action and Decision-making

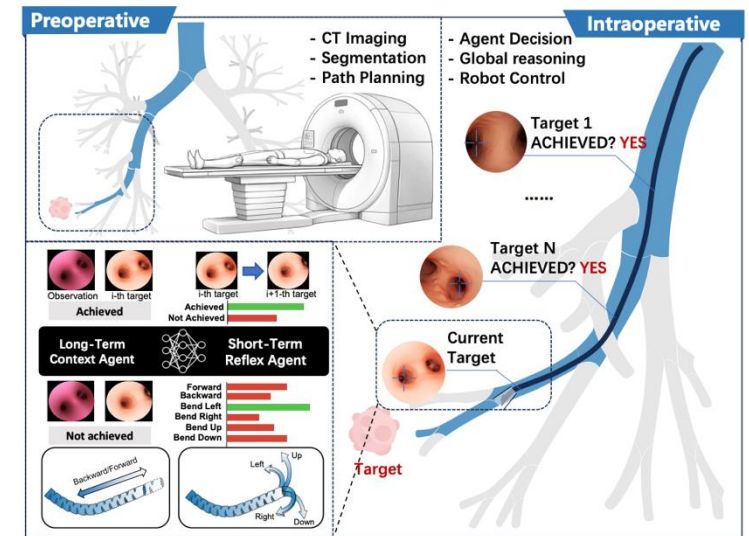
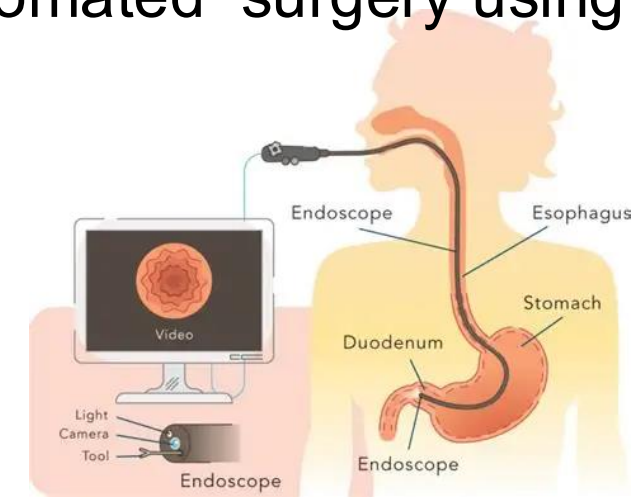


The field of Surgical Robotics and Today's Focus

Automated surgery (da Vinci surgical robot)



Automated surgery using Notes



Action and Decision-making for Intelligent Surgical Robotics: SuRResT (RAL 2026)

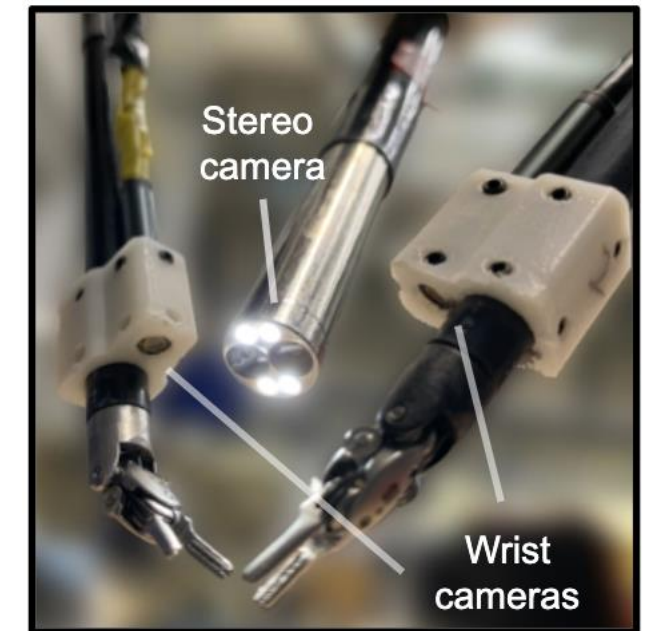
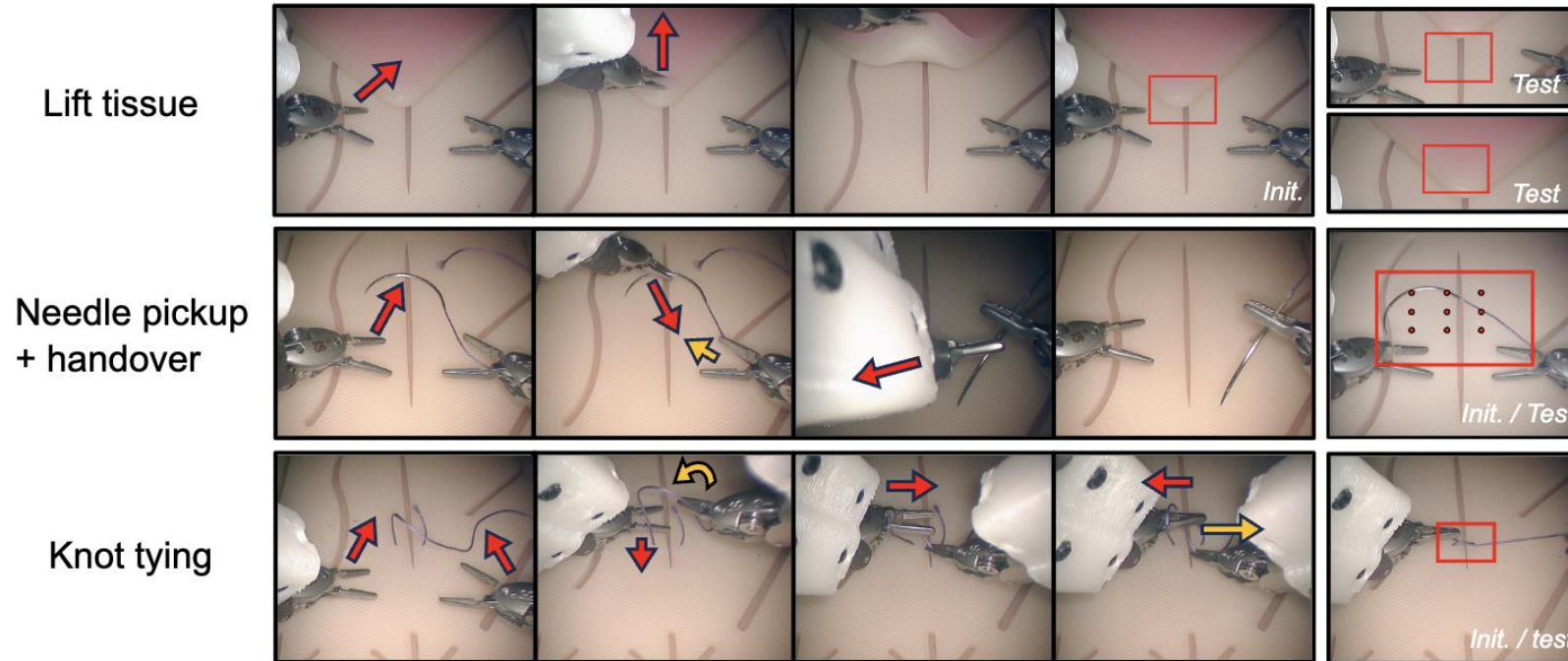


Some previous work has explored imitation learning in surgical robot, such as SRT(Surgical Robot Transformer)

However, imitation learning encounters critical challenges in clinical settings:

They uses two wrist cameras to provide extensive visual information, which might not be applicable in clinical settings.

Require over 200 demonstrations per task, a scale of data collection that is difficult to achieve in clinical practice.

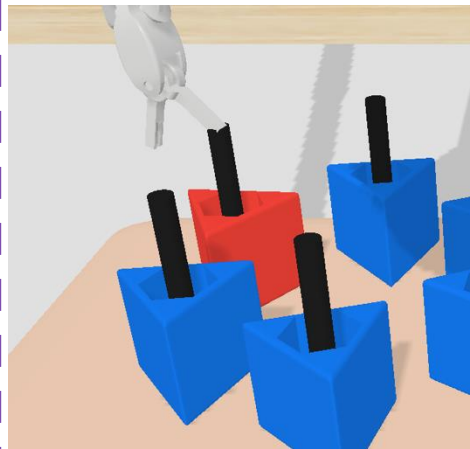


Action and Decision-making for Intelligent Surgical Robotics: SuRResT (RAL 2026)

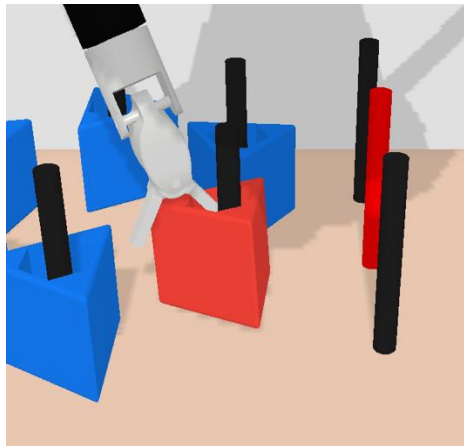


Precise manipulation is critical in surgical robotics, where sub-millimeter misalignments can lead to task failure or safety risks.

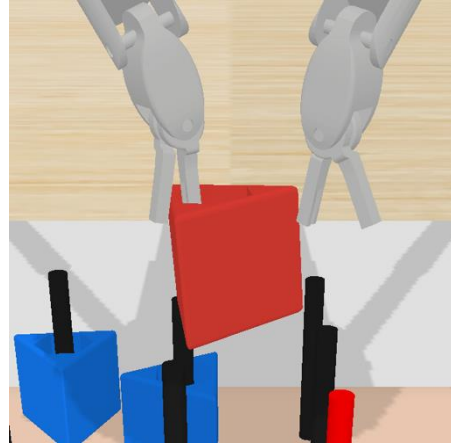
While imitation learning methods like Action Chunking Transformers (ACT) have shown promise in learning long-horizon behaviors, they often struggle with local precision — especially in placement or contact-intensive phases with only third-person-view RGB observations as in clinical settings (i.e., without wrist camera).



Collision with Peg

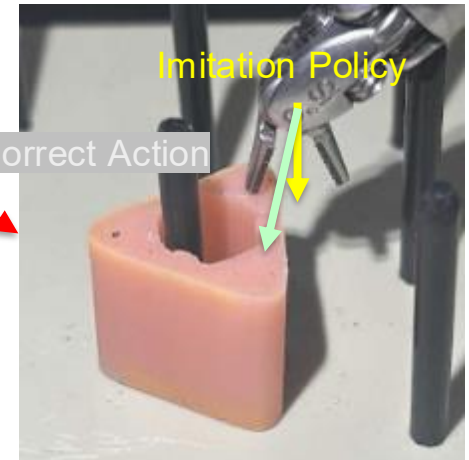
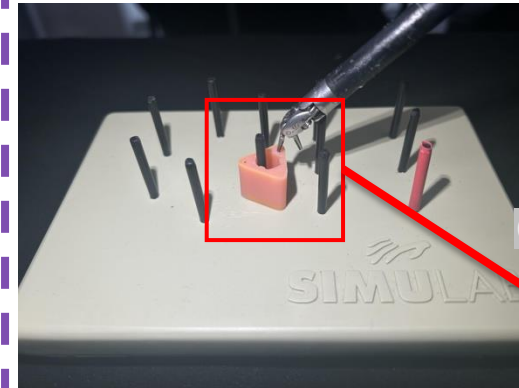


Incorrect Grasp Pose



Handover Misalignment

Simulation Environment



Real World Experiment

Action and Decision-making for Intelligent Surgical Robotics: SuRResT (RAL 2026)

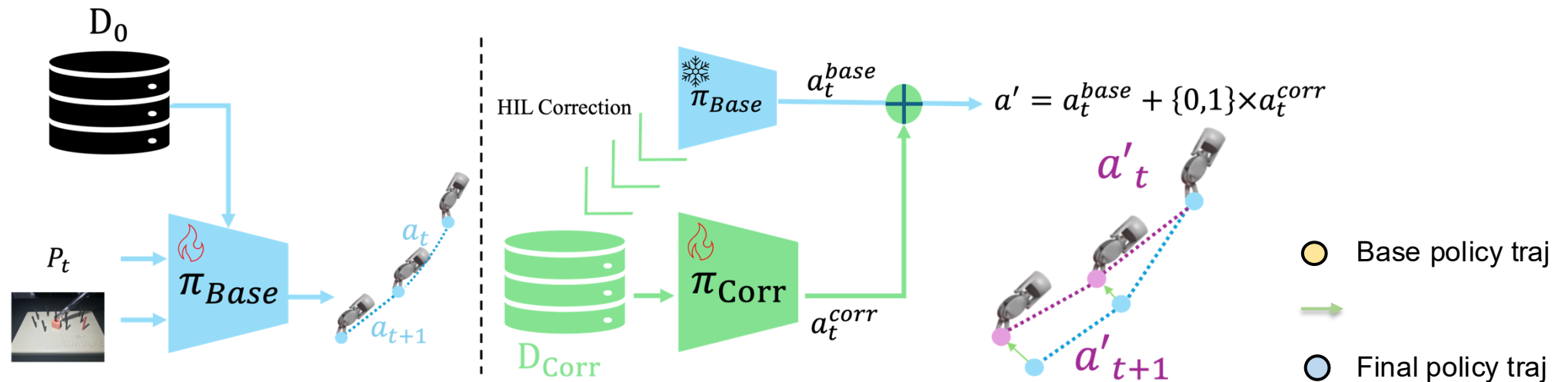


Our methods: Human in Loop - Residual Policy

We want to get rid of the wrist cameras and achieve comparable performances in surgical tasks.

We propose a residual correction policy that learns to adapt the base policy in real-time from online human corrections.

(A) Base Policy Learning: Using the ACT (Action Chunking Transformer) framework, the base policy is trained to predict short sequences of actions (chunks) from dataset with 'camera observations - action's paired dataset in simulation (SurRoL) and Real Robot (dVRK), enabling temporally coherent and structured decision-making.



Action and Decision-making for Intelligent Surgical Robotics: SuRResT (RAL 2026)

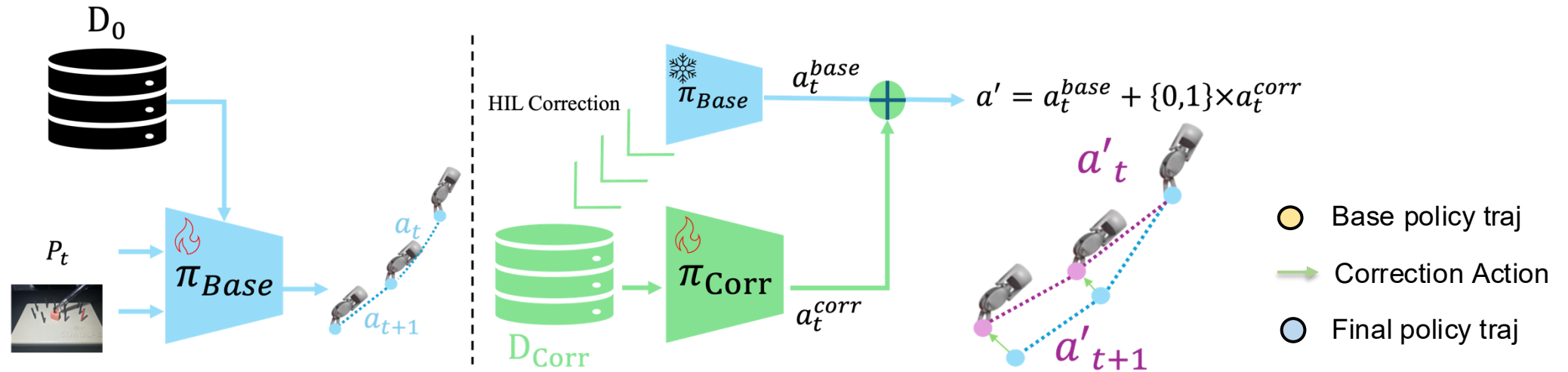


Our methods: Human in Loop - Residual Policy

(B) Policy Deployment: The base policy is deployed in simulation to perform surgical manipulation tasks; actions are executed unless a failure is detected.

Human-in-the-Loop Correction: When the base policy is about to fail, a human provides real-time corrections through the dVRK MTM interface. The correction data is then used to train the Residual Policy.

Once the residual policy is deployed, the final executed action will be a gated sum of the base policy and the correction action from residual policy.

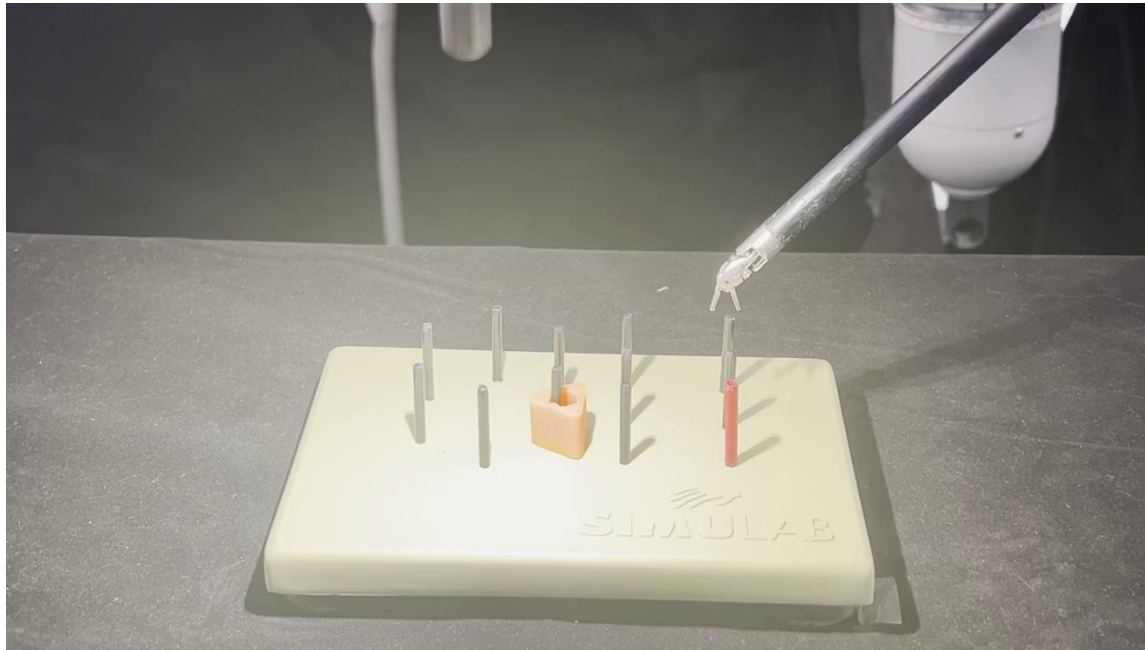


Action and Decision-making for Intelligent Surgical Robotics: SuRResT (RAL 2026)

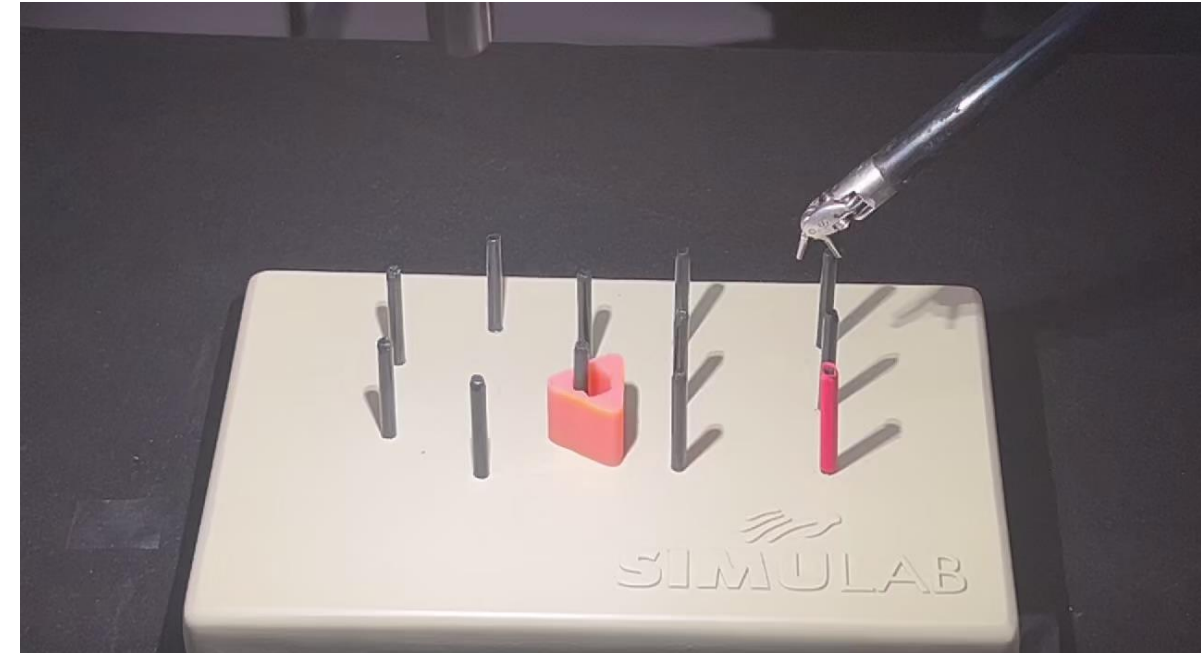


dVRK: With 50 demos to initialize the BC policy and 50 demos collected via human-in-loop correction.

Vanila BC (Base policy)



HIL+OOD Correction (ours)



Perception for Intelligent Surgical Robotics: SIA-RAPN

Surgical Strategy Prediction

双层缝合举例：

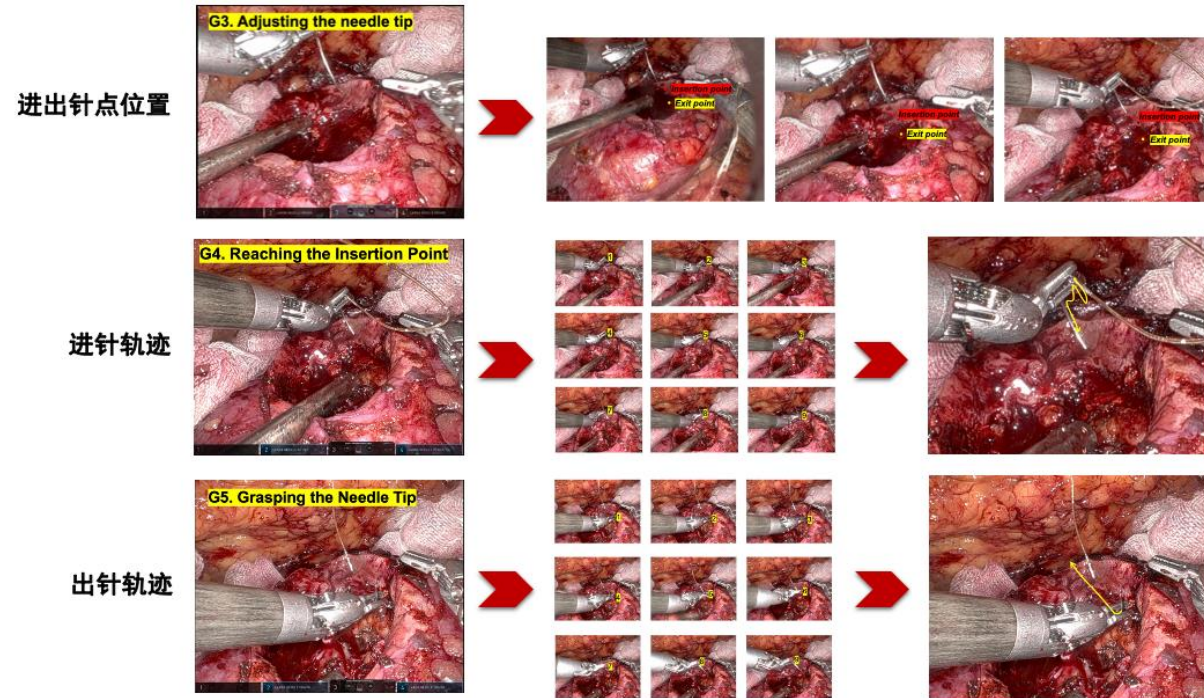


单层缝合举例：

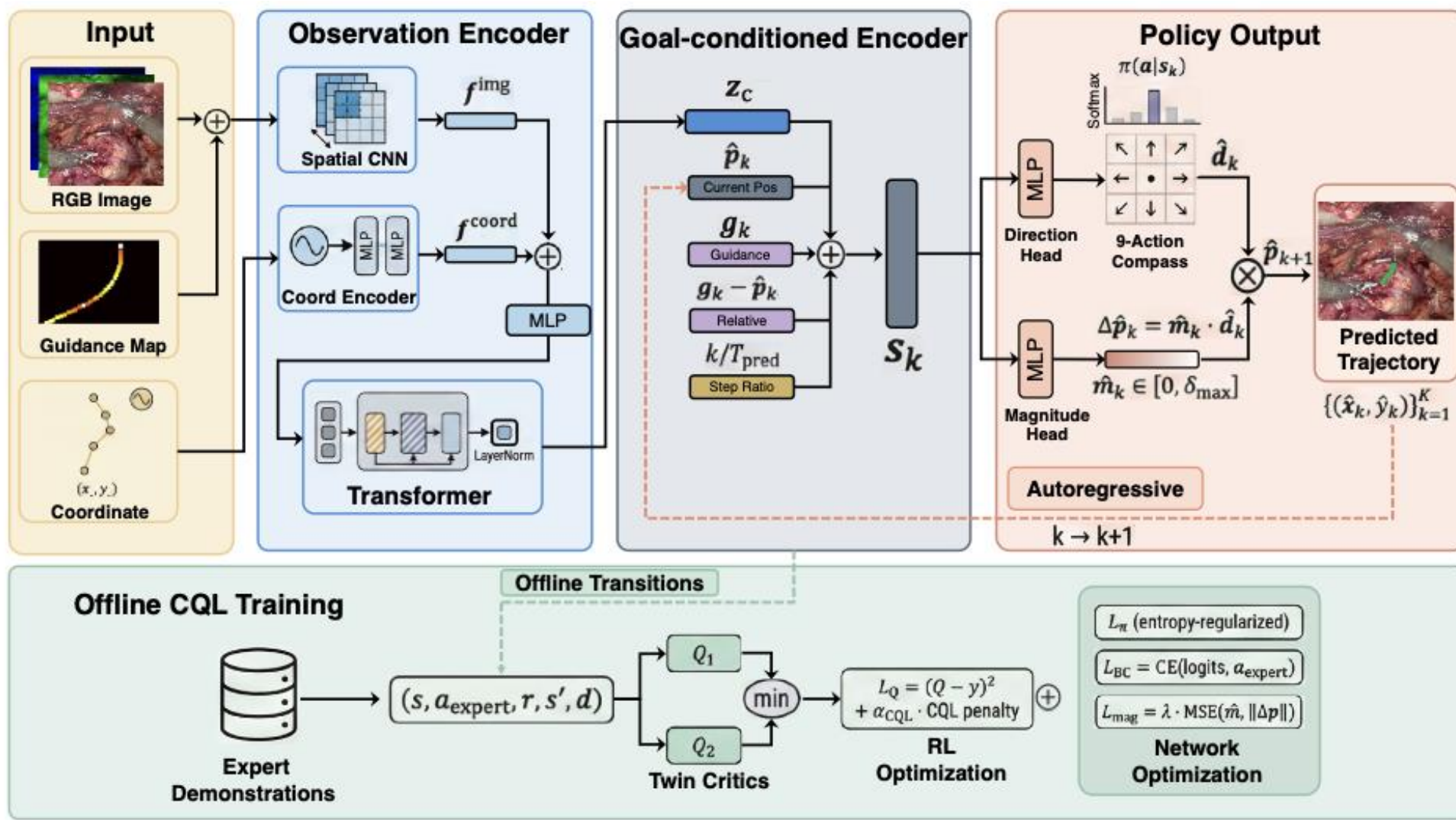


Perception for Intelligent Surgical Robotics: SIA-RAPN

Surgical Strategy Prediction



Perception for Intelligent Surgical Robotics: SutureFormer (MICCAI2026)



Perception for Intelligent Surgical Robotics: SutureFormer (MICCAI2026)



Method	Seed	Obs = 6, Pred = 3			Obs = 3, Pred = 6		
		ADE (↓)	FDE (↓)	FD (↓)	ADE (↓)	FDE (↓)	FD (↓)
VanillaCNN	–	208.79 ± 93.65	221.39 ± 101.68	234.47 ± 100.29	194.86 ± 84.86	227.73 ± 112.16	249.74 ± 101.67
BC [2]	–	128.15 ± 85.30	146.24 ± 100.87	156.83 ± 99.81	135.65 ± 75.23	182.31 ± 109.62	193.86 ± 104.57
GAIL [8]	–	268.31 ± 128.76	281.42 ± 136.55	301.98 ± 133.52	242.24 ± 117.38	268.84 ± 141.19	314.21 ± 126.19
IBC [6]	–	233.84 ± 132.22	250.76 ± 150.40	276.25 ± 141.66	214.63 ± 105.23	250.88 ± 153.55	288.38 ± 136.58
iDiff-IL [12]	–	179.65 ± 104.41	197.80 ± 115.62	208.52 ± 114.79	185.78 ± 95.33	228.86 ± 134.08	246.57 ± 123.23
CondDiff [12]	–	155.02 ± 90.48	176.03 ± 105.68	186.28 ± 104.52	158.20 ± 90.33	203.38 ± 128.04	219.98 ± 119.80
MID [7]	–	148.64 ± 93.01	165.28 ± 107.93	185.39 ± 105.36	154.65 ± 80.46	213.77 ± 128.01	225.18 ± 120.95
Ours	42	56.01 ± 32.81	85.51 ± 59.51	88.15 ± 57.21	93.22 ± 45.54	156.38 ± 89.90	160.75 ± 86.71
	123	55.54 ± 29.93	84.77 ± 51.15	85.67 ± 50.33	97.57 ± 49.59	167.84 ± 103.11	170.45 ± 101.23
	456	52.95 ± 29.73	80.22 ± 52.41	81.97 ± 50.98	96.51 ± 46.44	165.49 ± 99.19	170.64 ± 96.10
	789	55.11 ± 31.02	84.76 ± 54.17	85.82 ± 53.24	98.40 ± 46.65	165.96 ± 97.21	170.64 ± 93.95
	1024	58.54 ± 32.09	93.05 ± 54.67	93.68 ± 53.94	97.75 ± 45.25	167.29 ± 87.88	169.68 ± 86.63





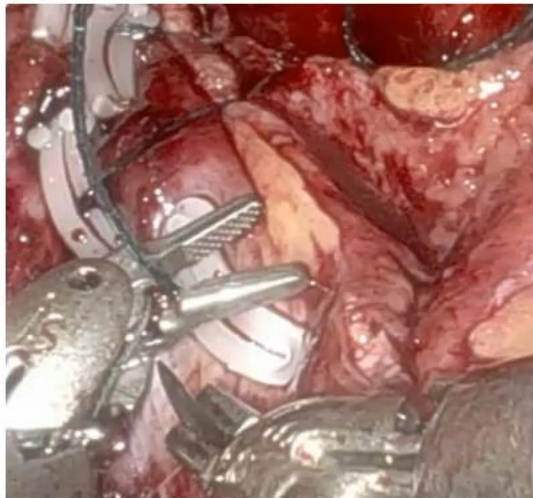
真实轨迹 vs 我们方法的预测轨迹



(1.a)



(1.b)



(1.c)

真实轨迹 vs 最好的基线方法的预测轨迹



(2.a)



(2.b)



(2.c)



病人1

成功案例

极端案例



(1.a)



(1.b)



(1.c)



(1.d)

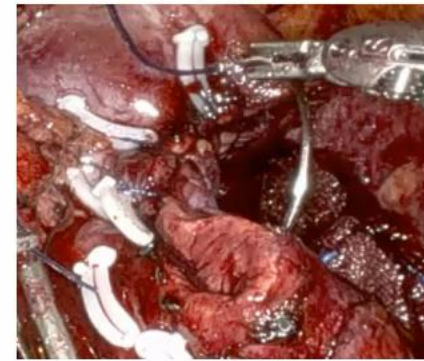
病人2



(2.a)



(2.b)

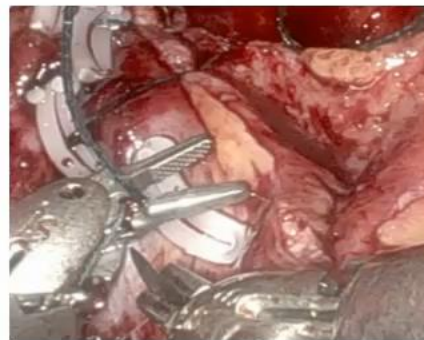


(2.c)

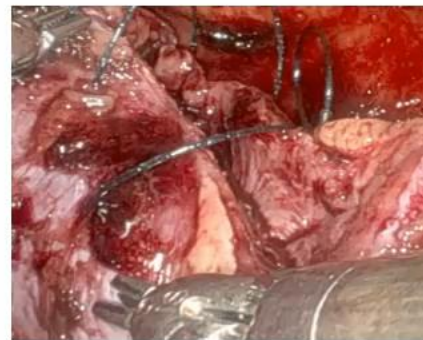


(2.d)

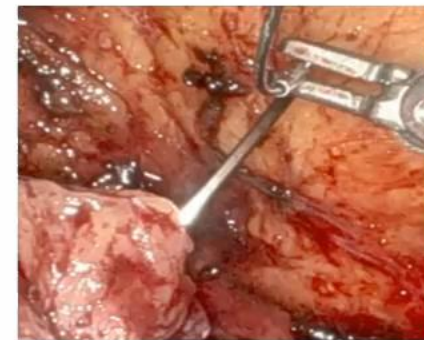
病人3



(3.a)



(3.b)



(3.c)



(3.d)

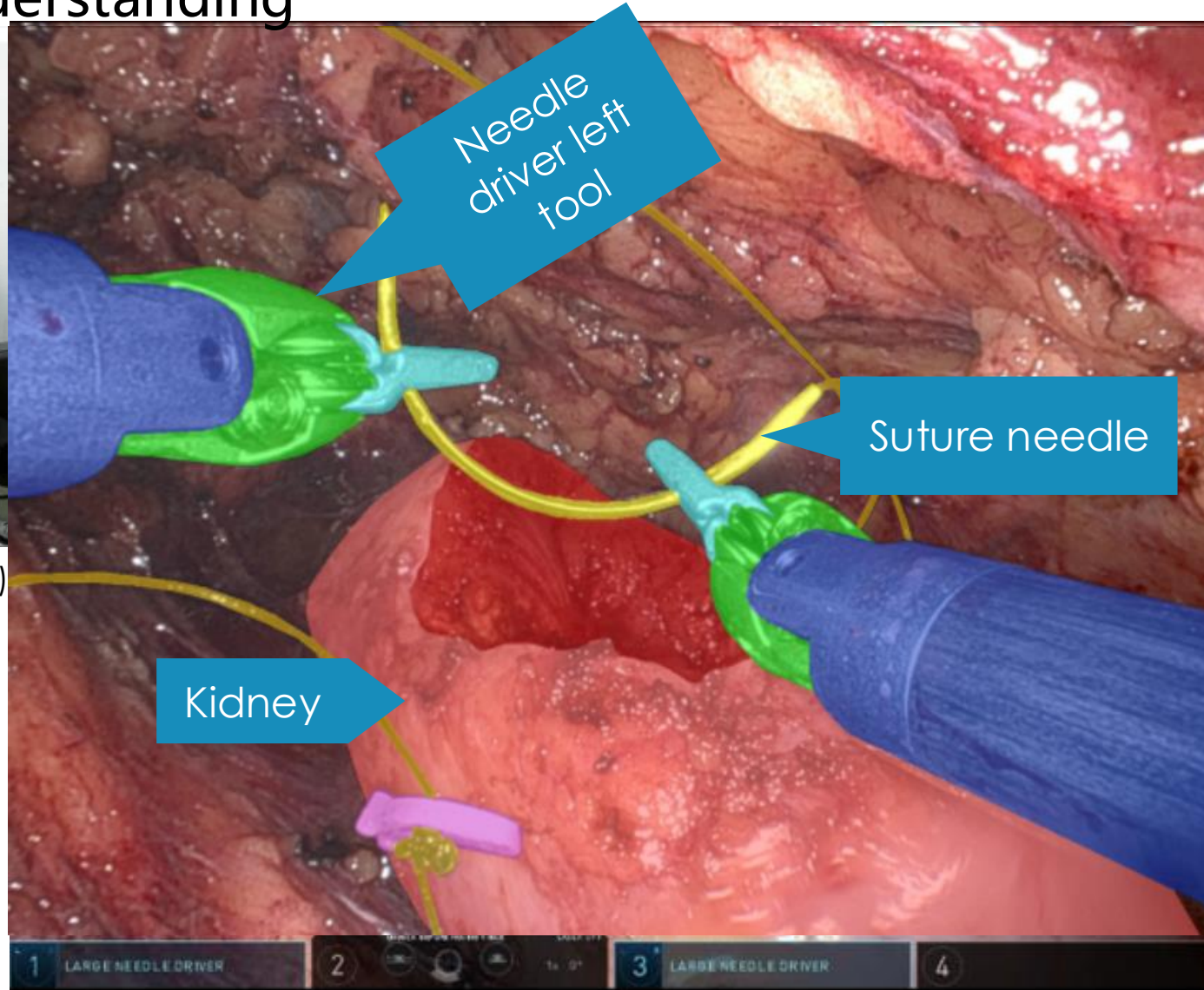
观测轨迹 真实轨迹 我们方法的预测轨迹 最好的基线方法的预测轨迹

Perception for Intelligent Surgical Robotics: SIA-RAPN

Surgical Scene Understanding



Live demonstrations (expertise)

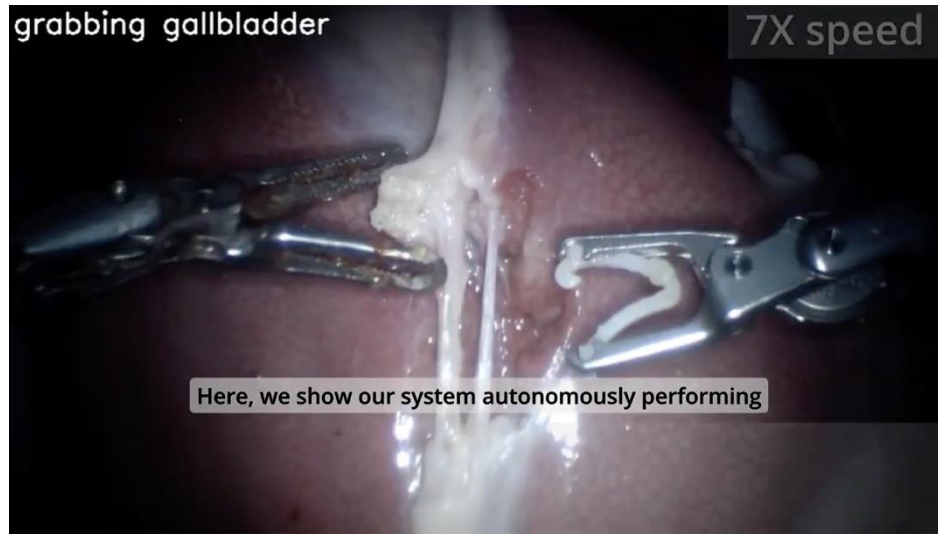


The sutures get all
scrambled and
are then completely
lost in the deep
left corner, entering
the robot's
medial arm not aiming
the eyes but the
lateral temporal
lobe, where they
are not visible in the
telemetry (live:
needle).

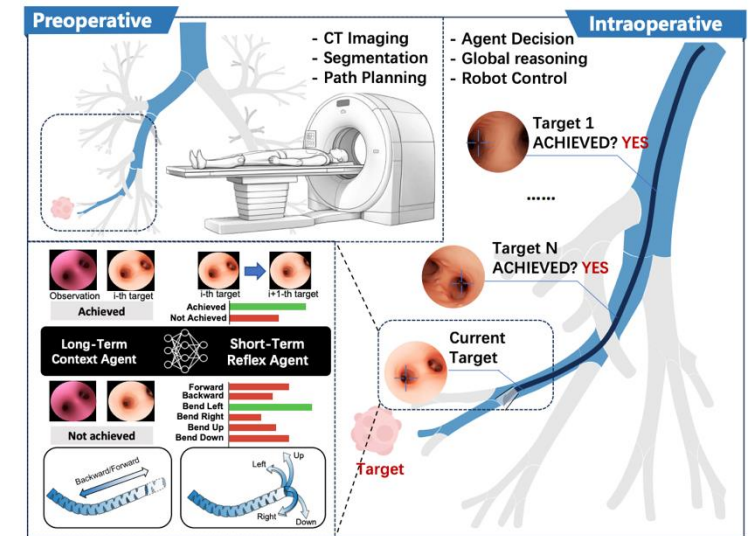
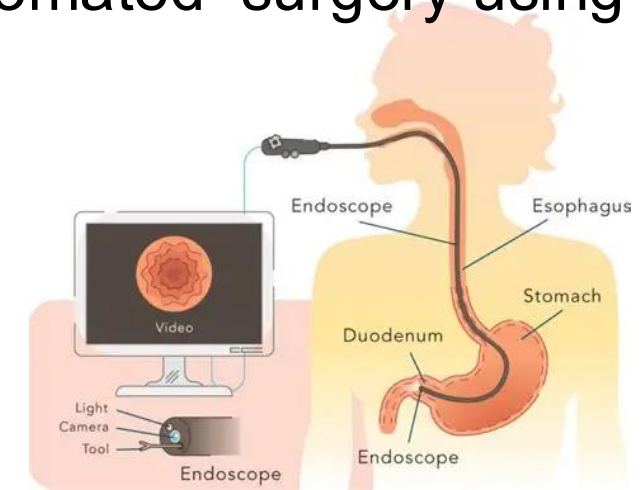
2. The surgical field is clear (V7: 0). The camera is still (V8: 0).

The field of Surgical Robotics and Today's Focus

Automated surgery (da Vinci surgical robot)



Automated surgery using Notes



Surgical Robotics Team at Mohism Lab



Peizhi Rong (PhD)

UG: Northeast Agricultural University
PG: Shandong University
Focus: Vary Stiffness Soft Robotics

Collaboration with Dr Chongjing Cao at SIAZ and Prof Gu at SJTU



Yuanyuan Liao

UG: Guangxi University of Technology
PG: University of Macau
Focus: Vary Stiffness Soft Robotics

Collaboration with Dr Chongjing Cao at SIAZ and Prof Gu at SJTU



Yan Bowen

UG: Chang' an University
PG: University of Macau
Focus: Vary Stiffness Soft Robotics (Smart control part)

Collaboration with Dr Chongjing Cao at SIAZ and Prof Gu at SJTU



Yubo Cao

UG: Shenyang Aerospace University
PG: University of Macau
Focus : Vary Stiffness Soft Robotics (Mechanical part)

Collaboration with Dr Chongjing Cao at SIAZ and Prof Gu at SJTU

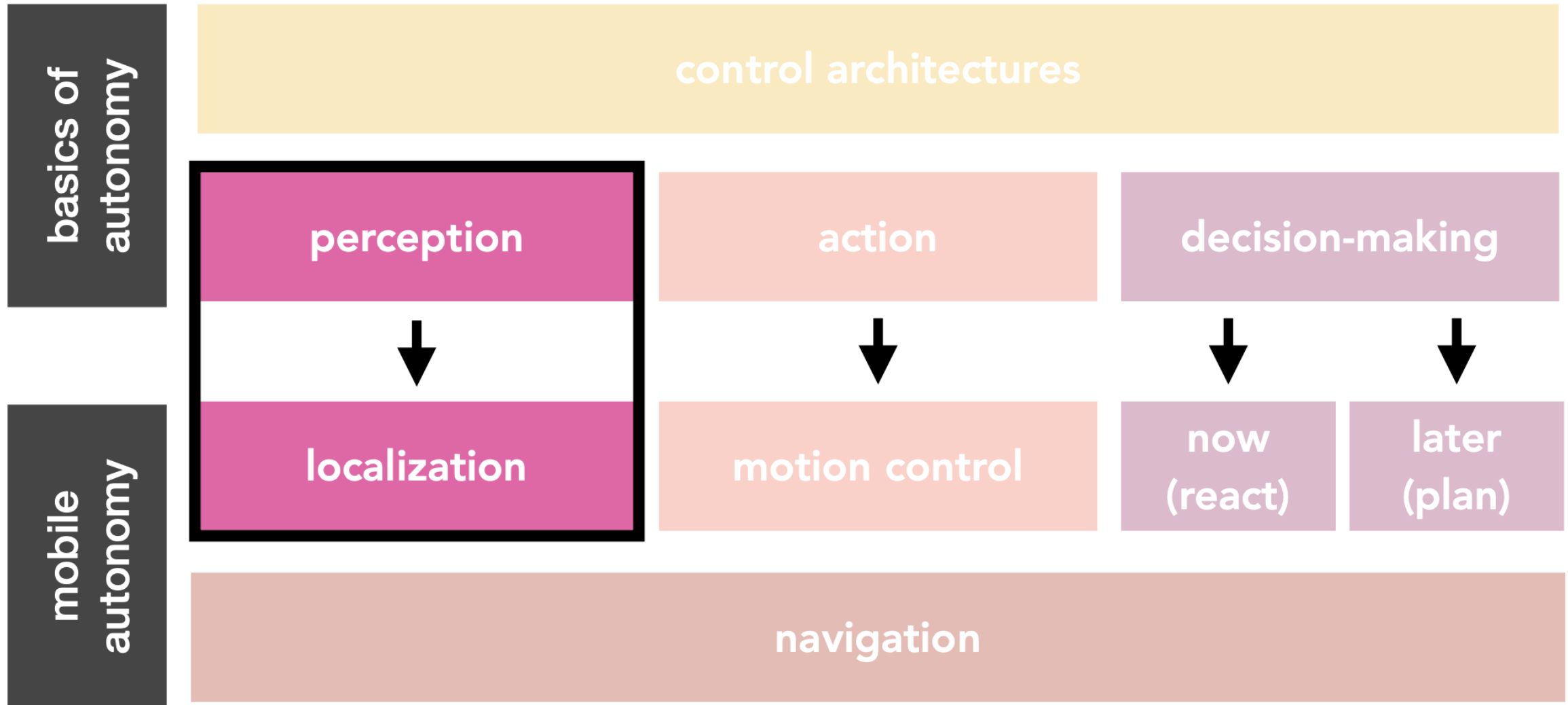


澳門大學
UNIVERSIDADE DE MACAU
UNIVERSITY OF MACAU



科技學院
Faculdade de Ciências e Tecnologia
Faculty of Science and Technology

Basics of Autonomy: Perception to Localization





Perception and Localization for Intelligent NOTES: M3-DEGREES Net (IEEE JBHI 2025)

► Problem solved:

- Conventional **monocular** endoscopic systems lack
 - **Reliable** metric **depth**,
 - **Limiting** safe **navigation**, lesion **localization**, instrument **control** in complex endoluminal surgeries

► Method:

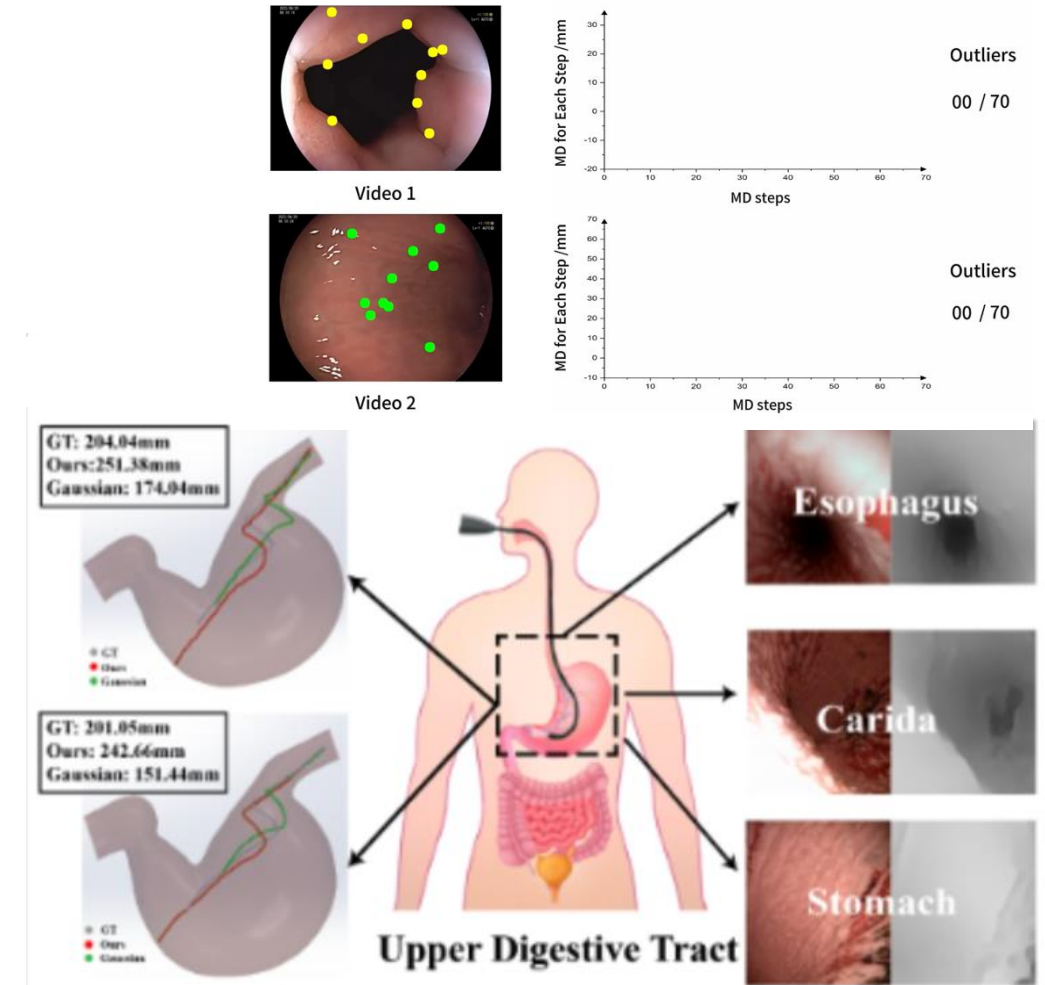
- Fuses **GAN depth** and **graph-based multi-modal** features to **estimate** metric **marching depth**.

► Results:

- Overall marching depth (MD) $\leq 27.3\%$, **13+ FPS**

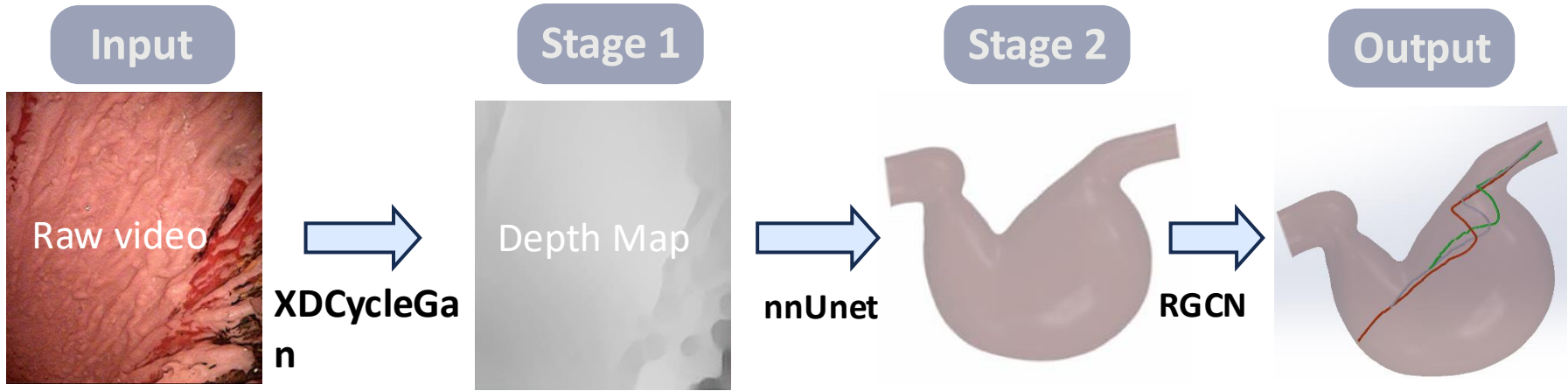
► Application scenarios:

- Gastrointestinal (GI) endoluminal Surgical navigation
- Robot-assisted Minimally invasive surgery (MIS)

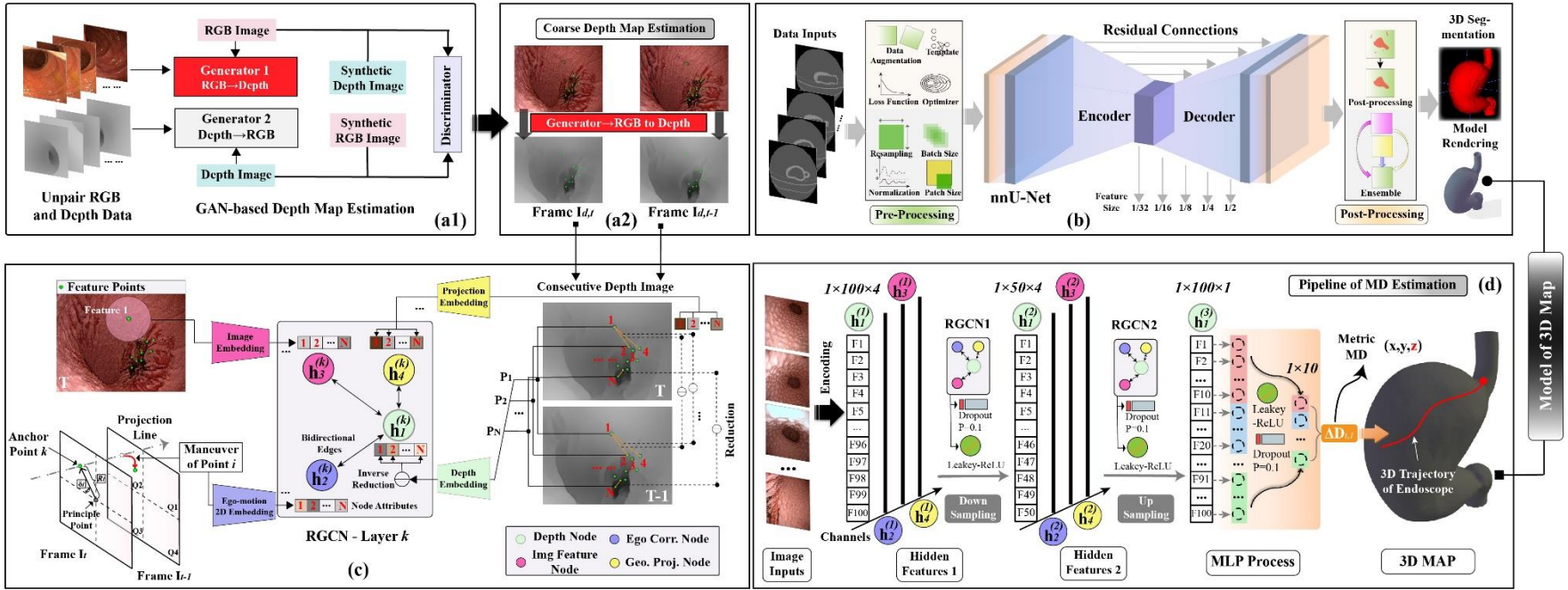
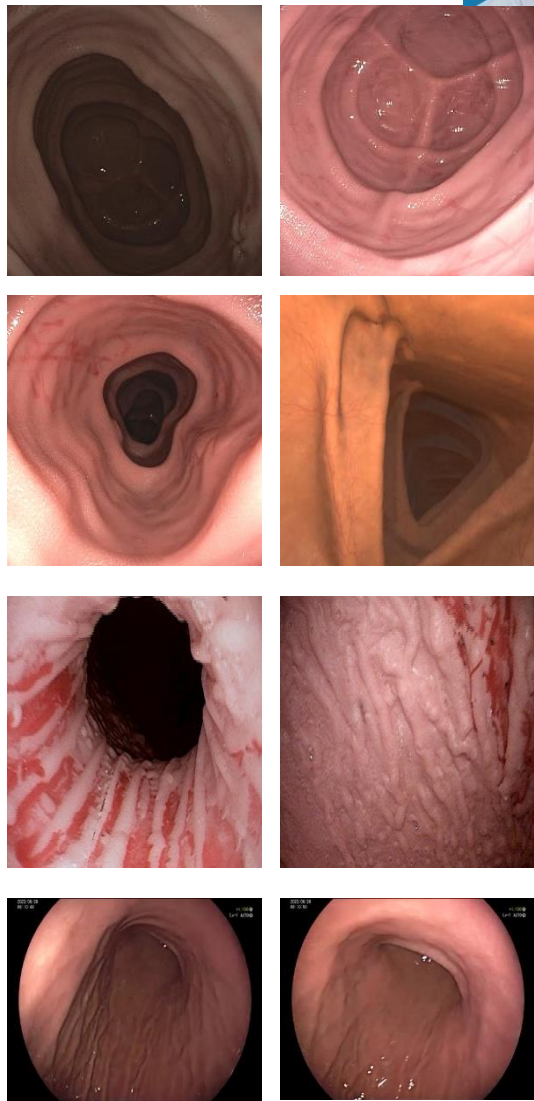




Perception and Localization for Intelligent NOTES: M3-DEGREES Net (IEEE JBHI 2025)



Multiple Datasets Tests





Perception and Localization for Intelligent NOTES: M3-DEGREES Net (IEEE JBHI 2025)

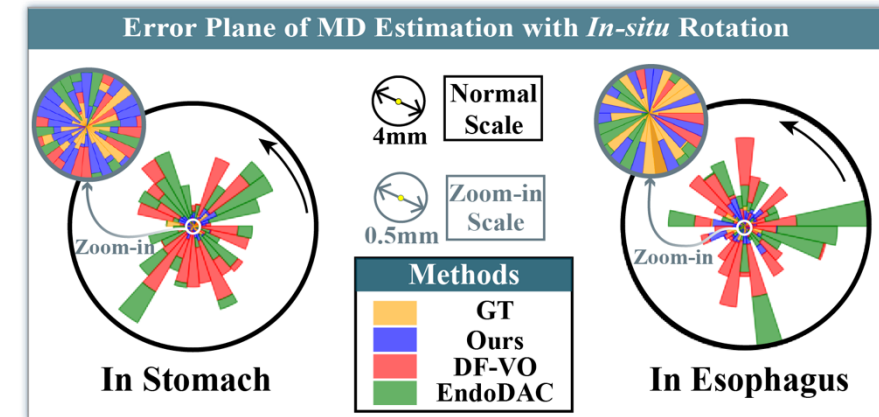
Quantitative Results of Metric Marching Depth Estimation under Esophagus and Stomach

QUANTITATIVE RESULTS CONCERNING THE AVERAGE STEP-WISE ESTIMATION ERROR e_p (STANDARD DEVIATION σ_p) OF MD ESTIMATION FOR IN-HOUSE DATASETS. THE BEST AND THE FIRST RUNNER-UP RESULTS ARE ACCORDINGLY HIGHLIGHTED IN BOLD FONT AND UNDERLINED.

Dataset	Method	Esophagus MD Range			Stomach MD Range		
		S-MD	M-MD	L-MD	S-MD	M-MD	L-MD
Camera 1	Monodepth2 [36]	107.6% (1.22)	89.1% (0.79)	<u>6.0% (0.06)</u>	134.8% (0.98)	105.9% (0.83)	77.5% (0.57)
	LiteMono [37]	86.1% (0.78)	61.4% (0.73)	53.4% (0.36)	103.2% (1.09)	82.2% (0.65)	58.2% (0.55)
	EndoSLAM [38]	<u>30.5% (0.18)</u>	24.8% (0.19)	32.3% (0.17)	120.8% (0.51)	101.1% (0.50)	114.8% (0.42)
	DF-VO [13]	32.8% (0.38)	28.1% (0.41)	23.2% (0.07)	65.6% (0.13)	65.8% (0.11)	71.5% (0.11)
	Depthanything2 [39]	51.8% (0.26)	48.7% (0.29)	52.1% (0.21)	103.4% (0.66)	86.3% (0.37)	77.3% (0.32)
	EndoDAC [40]	93.6% (0.54)	90.4% (0.53)	66.8% (0.38)	74.2% (0.40)	61.2% (0.26)	51.0% (0.37)
	GCN [33]	39.8% (0.57)	19.9% (0.44)	7.8% (0.05)	57.0% (0.46)	24.1% (0.22)	12.5% (0.09)
	Ours (M ³ -Net)	27.3% (0.34)	<u>20.1% (0.29)</u>	6.0% (0.06)	26.1% (0.15)	9.3% (0.07)	9.7% (0.11)
Camera 2	Monodepth2 [36]	67.6% (0.84)	84.1% (0.92)	49.4% (0.34)	90.6% (0.88)	60.7% (0.48)	41.3% (0.28)
	LiteMono [37]	88.1% (0.84)	50.7% (0.55)	52.8% (0.39)	<u>38.1% (0.31)</u>	42.5% (0.35)	37.5% (0.35)
	EndoSLAM [38]	36.7% (0.18)	52.7% (0.21)	43.0% (0.09)	<u>39.1% (0.27)</u>	55.5% (0.25)	51.9% (0.19)
	DF-VO [13]	<u>35.0% (0.21)</u>	45.8% (0.21)	33.9% (0.12)	56.8% (0.18)	67.3% (0.12)	64.0% (0.08)
	Depthanything2 [39]	206.3% (0.89)	39.5% (0.29)	25.4% (0.11)	153.9% (0.81)	33.6% (0.29)	30.5% (0.08)
	EndoDAC [40]	72.8% (0.47)	47.8% (0.38)	22.8% (0.25)	82.4% (0.34)	53.3% (0.35)	51.5% (0.30)
	GCN [33]	50.9% (0.45)	16.4% (0.25)	<u>14.4% (0.07)</u>	42.5% (0.37)	12.9% (0.10)	12.4% (0.15)
	Ours (M ³ -Net)	27.2% (0.22)	<u>21.1% (0.20)</u>	13.1% (0.11)	24.9% (0.20)	<u>16.2% (0.11)</u>	11.9% (0.10)

TABLE IV
NUMBERS OF ESTIMATION OUTLIERS IN ESOPHAGUS AND STOMACH.

Method	Camera 1		Camera 2	
	Esophagus	Stomach	Esophagus	Stomach
Monodepth2 [36]	17	32	23	17
LiteMono [37]	18	23	7	17
EndoSLAM [38]	2	39	13	0
DF-VO [13]	2	1	3	1
Depthanything2 [39]	1	25	9	3
EndoDAC [40]	5	5	0	1
GCN [33]	0	1	0	0
Ours(M ³ -Net)	0	0	0	0

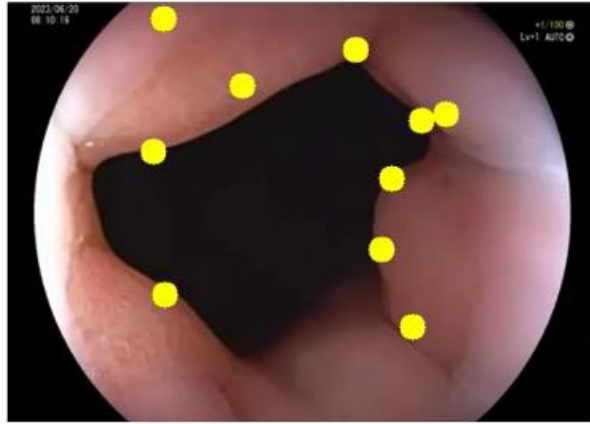


- Higher accuracy and robustness
- Higher adaptability of camera rotation

- Fewer outliers
- Less precision variation



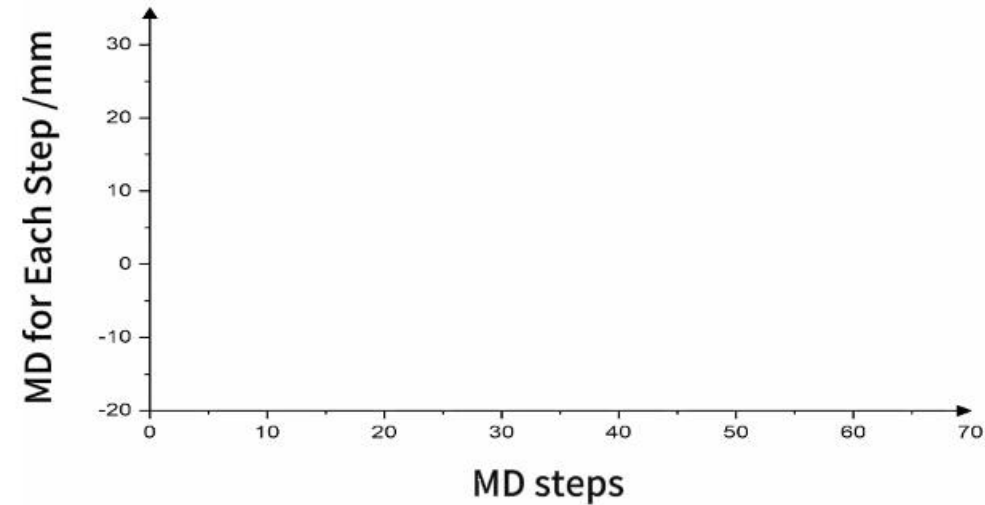
Perception and Localization for Intelligent NOTES: M3-DEGREES Net (IEEE JBHI 2025)



Video 1

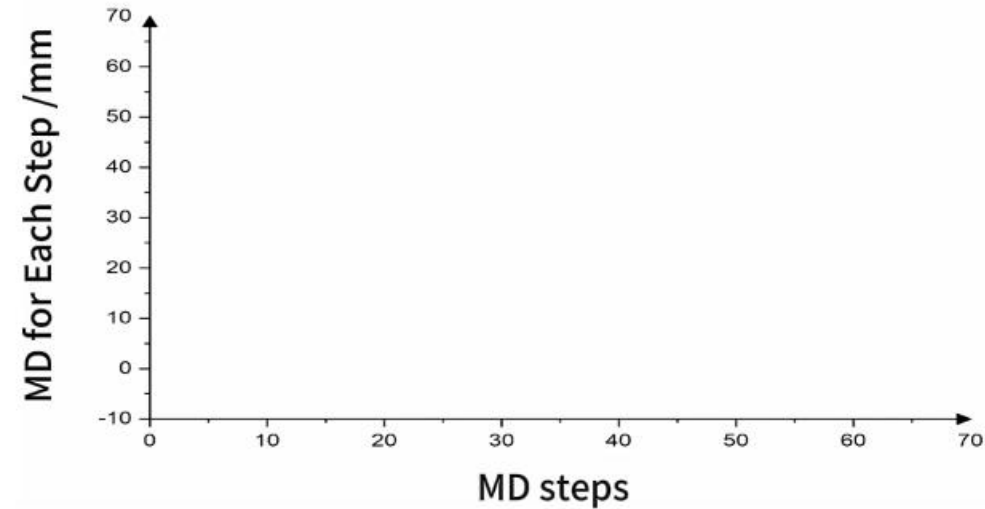


Video 2



Outliers

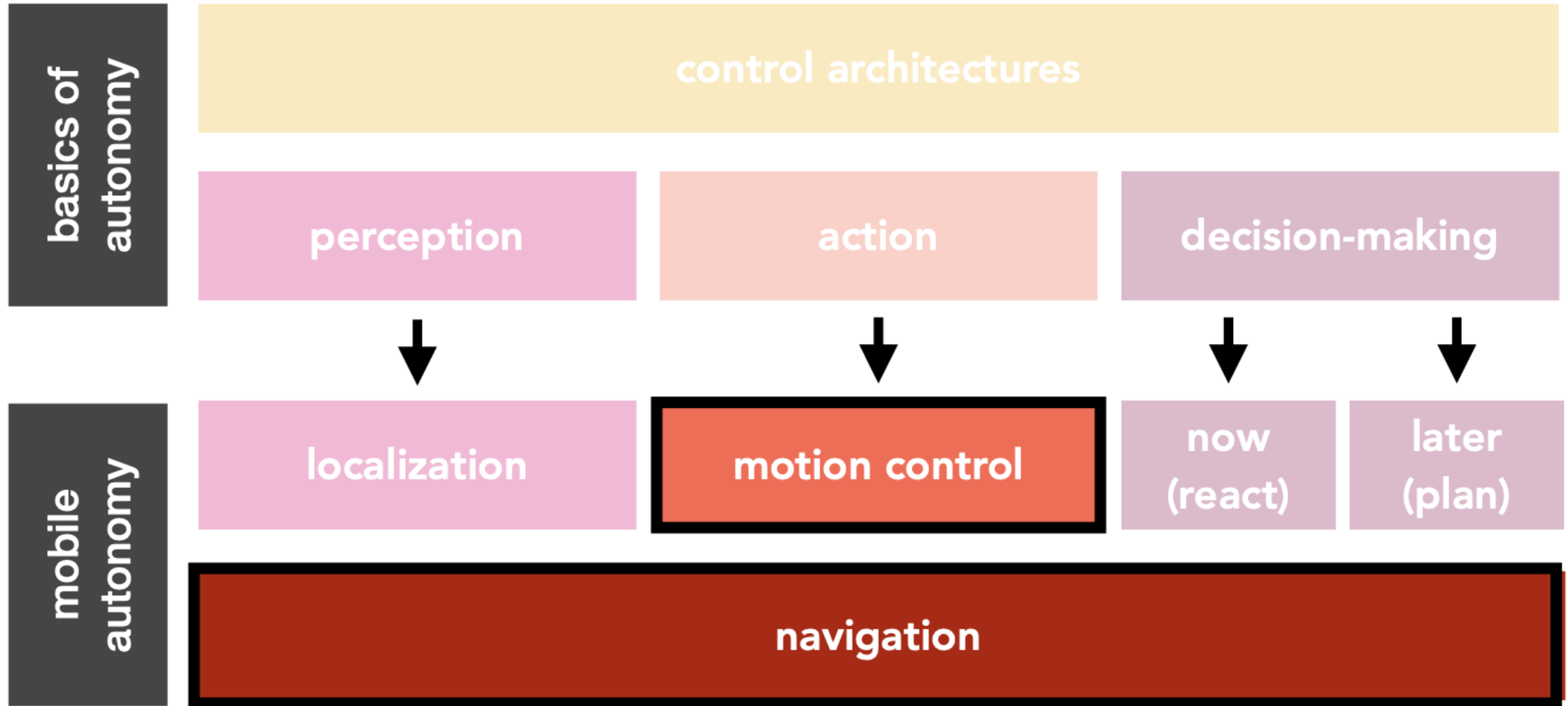
00 / 70



Outliers

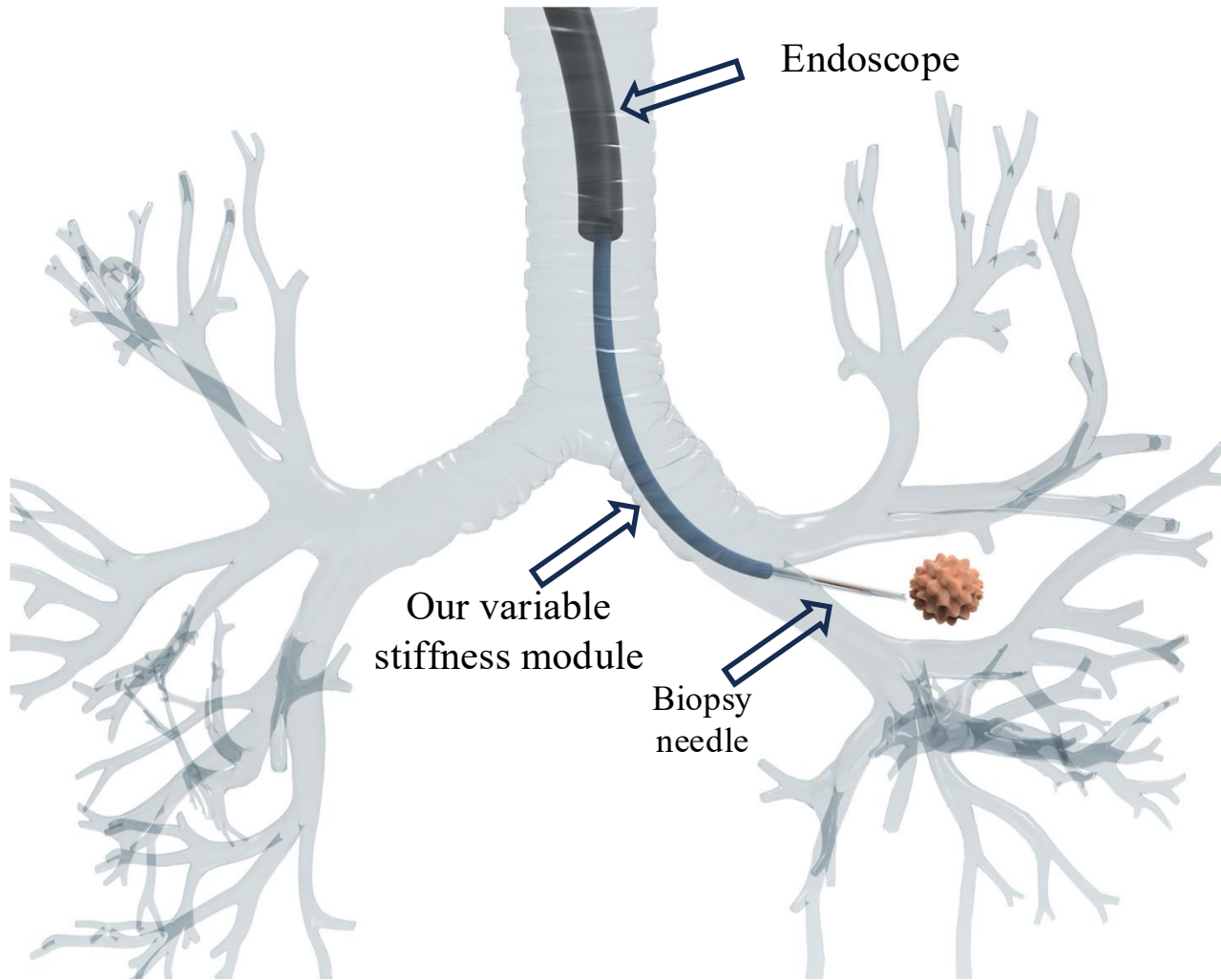
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Basics of Autonomy: Action



New Material to Control Intelligent Surgical Robotics:

From Variable Stiffness Mechanism (VSM) → Variable Stiffness Robot (VSR) for NoTEs

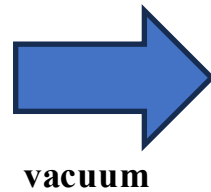
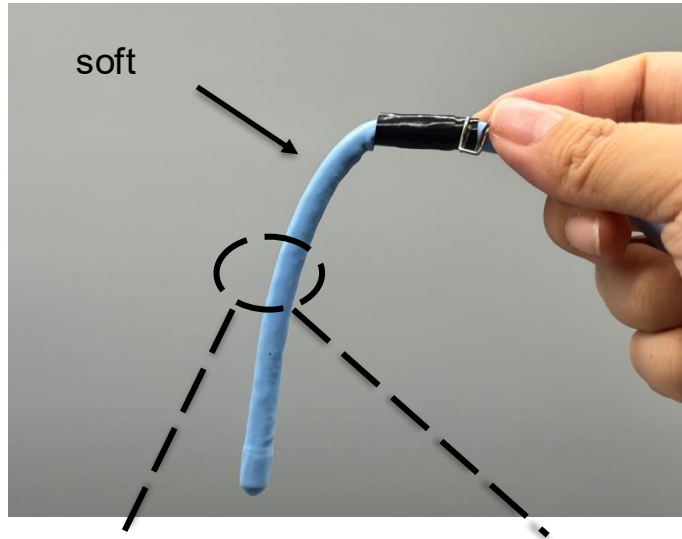


1. **Optimization** : stiffness change factor, response speed
2. **Fabrication** : biocompatibility, monolithic 3D printing
3. **Integration** : minimally invasive surgery, natural orifice transluminal endoscopic surgery (NOTES)

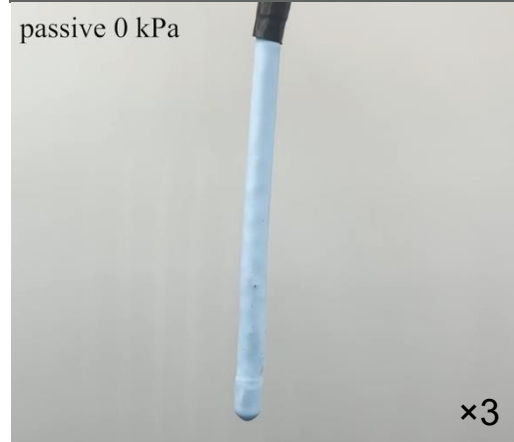
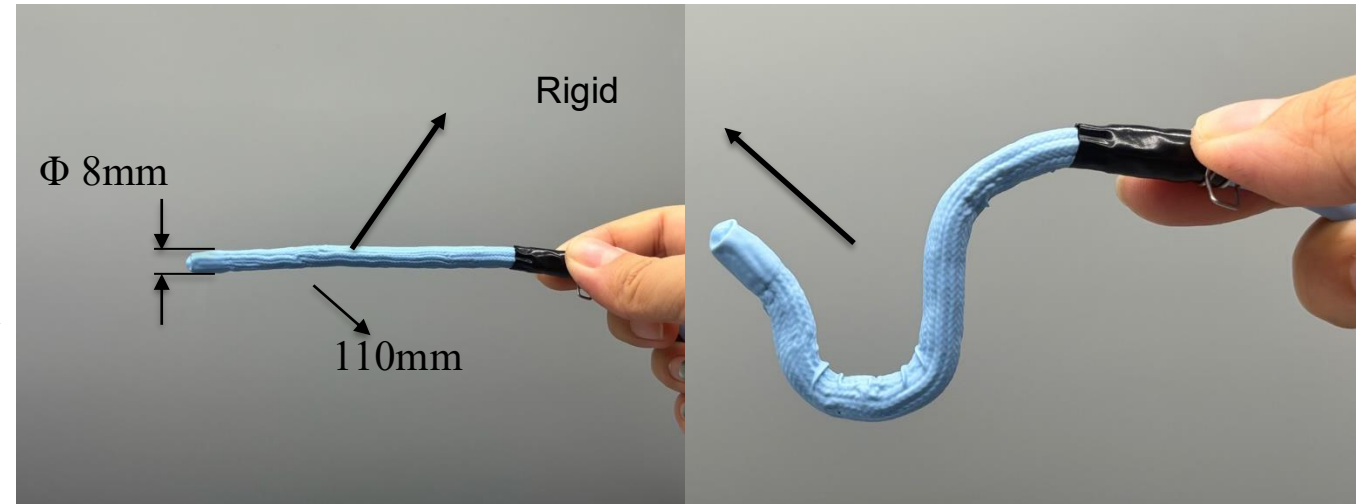
New Material to Control Intelligent Surgical Robotics: VSM (IROS 2025)



Passive 0 kPa



Active -80 kPa



Working principle

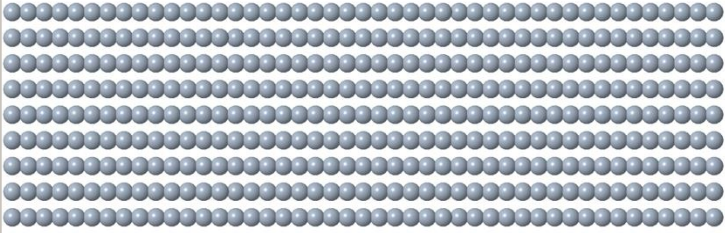
- Use a group of metal chains
- Soft when no pressure
- Rigid when at vacuum

New Material to Control Intelligent Surgical Robotics: VSM (IROS 2025)

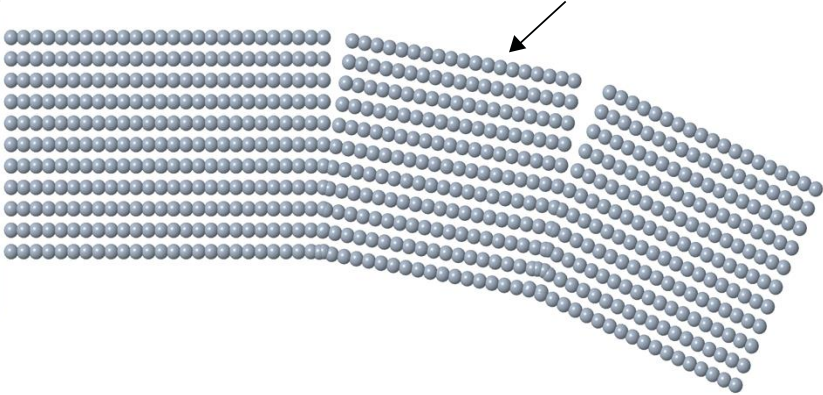


Conventional partical jamming

Under vacuum



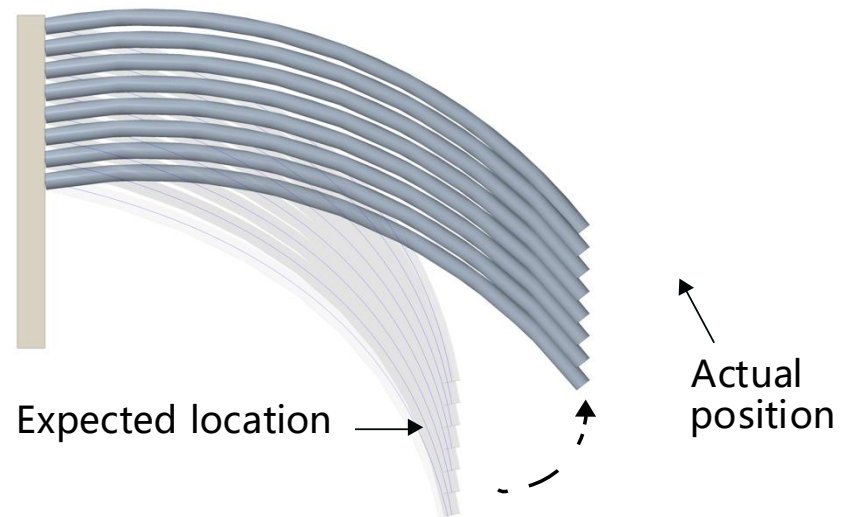
Bending will create gaps



- Particles in a vacuum distribute unevenly from loose to stacked, **creating voids** when bent.
- **Difficult to achieve arbitrary stiffness levels** with particle jamming.

Conventional fiber jamming

Under vacuum



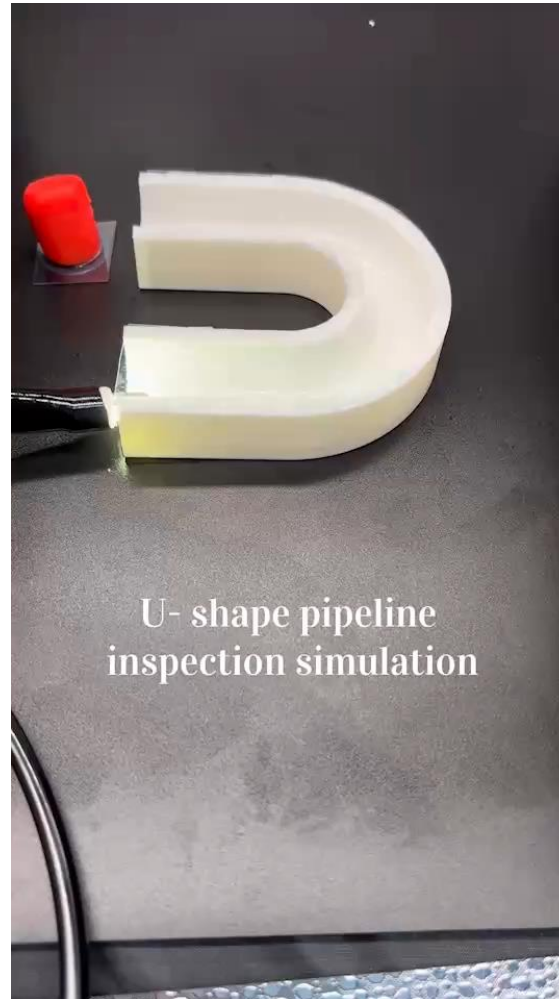
- Elastic fibers make the variable-stiffness structure **spring back** from its intended to its actual bent position.
- The stiffness change factor **is limited**.

New Material to Control Intelligent Surgical Robotics:

Application Scenario: Variable Stiffness Endoscopy



Simulate through
S-shaped pipeline

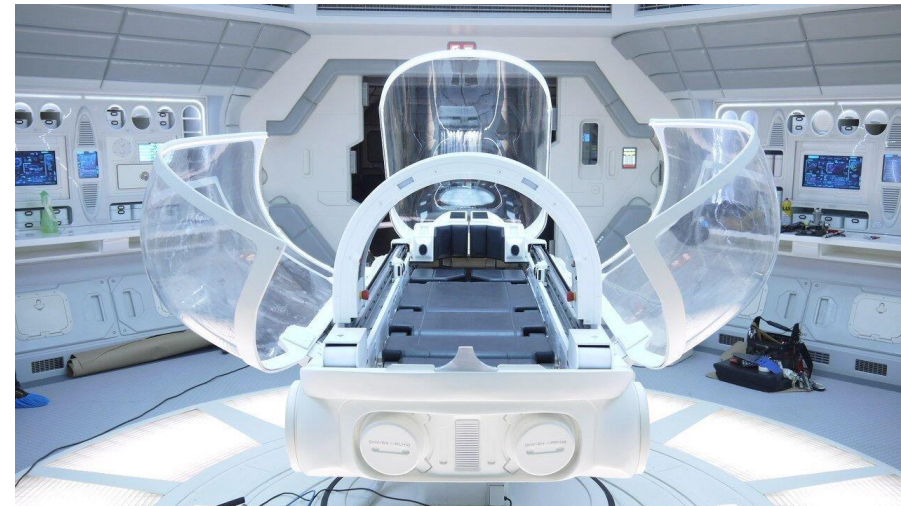


Simulate through
U-shaped pipeline



Force analysis of stretching biopsy forceps at
-80 kPa and 0 kPa.

Two dreams, One Goal



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Faculty of Science and Technology

Opening for Master Projects at Mohism Lab

Feel free to send me email from your edu email (recommended) with CV and your research interest to my work email: qingbiaoli@um.edu.mo

<https://forms.gle/ety6izFKpdyRZCcH6>

Look at the list of master project and book a meeting with me on Tuesday or Friday morning via following link.

<https://calendly.com/qingbiao-qli/interview-with-qingbiao-30min>

Before the meeting, prepare some slides so that you can introduce yourself in the meeting ;)



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Faculdade de Ciências e Tecnologia
Faculty of Science and Technology

Lab space of MohismLab

- E11-1049

University of Macau
(Hengqin main Campus)



Robotics Lab

12th Floor

Yuxin Building at Hengqin



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GPUServer

Equipped with

1. Dual CPU
 - AMD EPYC" 9334 Processor 32-Core 3.85GHz
2. 5 x Nvidia RTX 6000 Pro



High-performance workstations

Equipped with

- CPU – AMD Threadripper 7960X
1. Nvidia 5090



Data storage servers

Equipped with

- Dell r540 – 64 G Memory
1. HDD – 6*16 TB



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Thanks for Listening!



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